

# The Boundary Test Program

## Applying $Q = F \cdot \beta \cdot d^2 \cdot Z$ to Coherent Pattern Boundaries Across Scales

**Registry:** [\[HOWL-PHYS-4-2026\]](#)

**Series Path:** [\[HOWL-PHYS-1-2026\]](#) → [\[HOWL-MATH-1-2026\]](#) → [\[HOWL-PHYS-3-2026\]](#) → [\[HOWL-PHYS-4-2026\]](#)

**DOI:** 10.5281/zenodo.zzz

**Date:** March 2026

**Domain:** Foundational Physics / Measurement Theory

**Status:** Complete

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## I. ABSTRACT

This paper defines a testing program that connects three prior results: the coherent pattern boundaries and anomaly correlations of [\[HOWL-PHYS-1-2026\]](#), the geometric invariant  $\beta = \pi / 4$  and unified equation  $Q = F \cdot \beta \cdot d^2 \cdot Z$  of [\[HOWL-MATH-1-2026\]](#), and the experimental gap in G measurements across Earth's Hill sphere documented in [\[HOWL-PHYS-3-2026\]](#).

The program asks one question: does the  $Q = F \cdot \beta \cdot d^2 \cdot Z$  framework, proven for geometric cross-sections across nine domains, apply to the coherent pattern boundaries where measurement anomalies have been documented?

The paper does not answer this question. It classifies which boundaries are geometric and which are not. It specifies seven tests ordered from immediately achievable to far future. It identifies three tests that are reanalyses of existing published data requiring no new experiments. It commits to a kill switch: if all three existing-data tests produce null results, the remaining tests are not motivated and the program stops.

One test — the Hubble tension cumulative transit calculation — is identified as underspecified. The per-boundary transit correction magnitude is not provided by the prior papers and has not been derived. This missing piece is stated as a primary limitation. The test cannot be executed until the per-transit correction is established by theoretical work this paper does not attempt.

Every premise is from the institution's own published literature or from the prior papers in this series. No new physics is proposed. No new mathematics is introduced. The

contribution is the specification of a test program connecting established results that have not been connected.

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## II. FOUNDATIONS

### 2.1 What MATH-1 Proved

[[HOWL-MATH-1-2026](#)] proved that nine equations across nine domains of applied mathematics share a common structure:

$$Q = F \cdot \beta \cdot d^2 \cdot Z$$

where  $\beta = \pi/4$  is the ratio of circular cross-sectional area to rectilinear bounding area. The proof is algebraic identity — each equation is algebraically identical to its standard form. The nine domains are geometry, pipe flow, drag, orifice flow, capacitance, electromagnetic aperture, antenna theory, beam optics, and thermal radiation. Every instance involves a circular or elliptical geometry interfacing with a rectilinear measurement frame. The geometric factor  $\beta \cdot d^2$  is invariant across all nine domains.  $F$  (driving term) and  $Z$  (domain-specific impedance) vary.

The scope of the proof: geometric cross-sections. Circular and elliptical areas interfacing with rectilinear measurement. The proof has not been shown to apply to non-geometric boundaries.

### 2.2 What PHYS-1 Identified

[[HOWL-PHYS-1-2026](#)] identified coherent pattern boundaries at multiple scales — electron vacuum polarization clouds, hadron confinement boundaries, proton internal structure, atomic shells, planetary gravitational wells, stellar boundaries, galactic structure, cosmological structure. Interior and exterior readings differ across these boundaries. Three measurement anomalies — the Hubble tension, the proton radius puzzle, and the muon  $g-2$  discrepancy — correlate with boundary transit count or interaction depth.

The scope of the identification: qualitative. The anomaly correlation is a pattern across three data points. No mathematical framework was applied. The correlation warrants investigation but does not constitute proof.

### 2.3 What PHYS-3 Established

[[HOWL-PHYS-3-2026](#)] established that every direct measurement of the gravitational constant  $G$  in the 227-year experimental record has been performed inside Earth's Hill sphere — at 0.45% or less of the Hill sphere distance. The persistent disagreement between  $G$  measurements from different experimental groups has not been resolved by two centuries of systematic error investigation. PHYS-3 proposed a direct  $G$  measurement at the L1 or L2 Lagrange point as the first boundary-crossing test. PHYS-3 also documented that every coupling whose boundary has been identified and whose running has been measured ( $\alpha$ ,  $\alpha_s$ ) shows a consistent single curve with no method disagreement,

while G — the one coupling whose boundary has not been identified — shows persistent method disagreement.

The scope of the establishment: the experimental gap is factual. The interpretation — that G may be boundary-dependent — is a hypothesis awaiting the proposed test.

## 2.4 What Has Not Been Connected

MATH-1 provides a framework proven for geometric cross-sections. PHYS-1 catalogs boundaries where anomalies correlate with boundary structure. PHYS-3 identifies a specific untested boundary for a specific measurement. These three results have not been connected. Whether the geometric framework applies to the physical boundaries is an open question. The answer might be no. This paper defines the tests that would determine the answer.

## III. WHICH BOUNDARIES ARE GEOMETRIC?

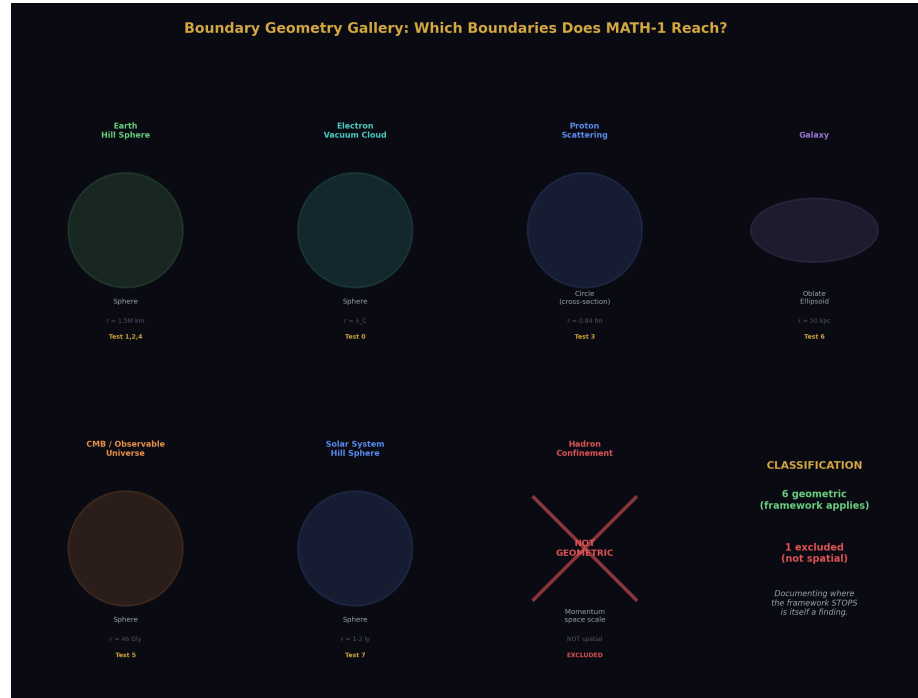


Figure 1: Fig. 1: Six boundaries classified as geometric (sphere, ellipse, scattering cross-section) and one excluded (confinement — not spatial). The framework's scope is visible.

### 3.1 The Criterion

MATH-1's framework applies to boundaries that are spatial geometries — circles, spheres, or ellipses — interfacing with a rectilinear measurement frame. Not all boundaries cataloged in PHYS-1 meet this criterion. Each boundary is evaluated individually below.

### 3.2 Earth's Hill Sphere

The Hill sphere is a sphere. It has a calculable radius (~1.5 million km). Its cross-section is circular. A measurement probe approaching Earth from outside encounters a spherical boundary between the solar reference frame — in which Earth is a point mass — and Earth's coherent gravitational interior where Earth's structure dominates.

The geometry is spatial and circular. The interface is between an interior frame and an exterior frame. The MATH-1 framework requires exactly this: a circular geometry interfacing with a rectilinear measurement frame.

**Classification: geometric. Framework applies in principle.**

### 3.3 Solar System Hill Sphere

Same structure as Earth's Hill sphere at a larger scale. The solar system's Hill sphere is a sphere with radius ~1-2 light years. The interface is between the Sun's coherent gravitational domain and the galactic field. Spatial, spherical, circular cross-section.

**Classification: geometric. Framework applies in principle. Untestable with current technology — no instruments at this boundary.**

### 3.4 Proton

The proton is a quantum object whose charge radius is defined operationally through scattering cross-sections. The electron-proton and muon-proton scattering cross-sections are circular — they are literally  $\pi r^2$  terms at each probe energy, measured in the center-of-mass frame. The probe interacts with the proton at a characteristic radius determined by the probe's energy and mass. The cross-section at each radius is circular.

The proton is not a classical sphere. But the measurement of the proton — the scattering cross-section — is a circular geometric cross-section in the precise sense MATH-1 requires. The framework applies to the measurement geometry, not to the proton's internal ontology.

**Classification: geometric at the level of measurement cross-sections. Framework applies in principle. The proton radius puzzle is the test case.**

### 3.5 Electron Vacuum Polarization Cloud

The virtual particle cloud surrounding a bare electron charge is spherically symmetric. The boundary between bare charge ( $\alpha \approx 1/127$  at high energy) and screened charge ( $\alpha$

$\approx 1/137$  at low energy) is a sphere at the characteristic Compton wavelength scale of the electron ( $\sim 3.86 \times 10^{-13}$  m). The measurement of  $\alpha$  at different probe energies involves scattering cross-sections — circular geometric cross-sections at each energy.

**Classification: geometric at the level of scattering cross-sections. The running of  $\alpha$  is the established measurement across this boundary. This is the calibration case — the known result the framework must reproduce before being applied to unknown cases.**

### 3.6 Hadron Confinement Boundary

Confinement is a property of QCD. The boundary where asymptotic freedom gives way to confinement is defined in momentum space — it is a scale in the running coupling constant, not a geometric surface in position space. There is no spatial sphere. There is no circular cross-section interfacing with a rectilinear frame in the sense MATH-1 requires.

**Classification: not geometric in the MATH-1 sense. The framework does not apply directly. This boundary is outside scope.**

This is not a failure. It is a scope finding. The framework applies to spatial geometric boundaries. Confinement is not a spatial geometric boundary. Documenting where the framework stops is as important as documenting where it applies.

### 3.7 Galaxy

The galactic disk is not spherical. It is oblate — roughly elliptical in cross-section when viewed edge-on. MATH-1 Section III establishes that  $\beta$  applies to all ellipses: the area of any ellipse equals  $\beta$  times its bounding rectangle, at all eccentricities.

The galactic boundary — where the galaxy's coherent gravitational structure gives way to the intergalactic field — is approximately elliptical.

**Classification: geometric with the ellipse generalization. Framework applies in principle. Calculationally difficult — the galactic boundary is not a clean ellipse and the rectilinear measurement frame is not obvious at galactic scales.**

### 3.8 Observable Universe / CMB Boundary

The observable universe is approximately spherical. The cosmic microwave background is emitted from the surface of last scattering — a spherical surface. Light transiting from the CMB to our instruments crosses this spherical boundary plus every coherent gravitational structure between the CMB surface and our instruments.

**Classification: geometric. The Hubble tension is the test case. The most complex and currently underspecified calculation in the program.**

### 3.9 Summary

| Boundary                  | Geometry                              | Spatial? | Classification                | Test Case                         |
|---------------------------|---------------------------------------|----------|-------------------------------|-----------------------------------|
| Earth Hill sphere         | Sphere                                | Yes      | Geometric                     | G measurement                     |
| Solar system Hill sphere  | Sphere                                | Yes      | Geometric                     | Not testable now                  |
| Proton                    | Measurement cross-section is circular | Yes      | Geometric                     | Proton radius puzzle              |
| Electron vacuum cloud     | Sphere / scattering cross-section     | Yes      | Geometric                     | Running of $\alpha$ (calibration) |
| Hadron confinement        | Momentum-space scale                  | No       | Not geometric — outside scope | —                                 |
| Galaxy                    | Oblate ellipsoid                      | Yes      | Geometric (ellipse)           | Rotation curves                   |
| Observable universe / CMB | Sphere                                | Yes      | Geometric                     | Hubble tension                    |

Six of seven boundaries are geometric. One is not. The framework's scope covers six boundaries. The exclusion is documented.

## IV. CALIBRATION: THE RUNNING OF $\alpha$

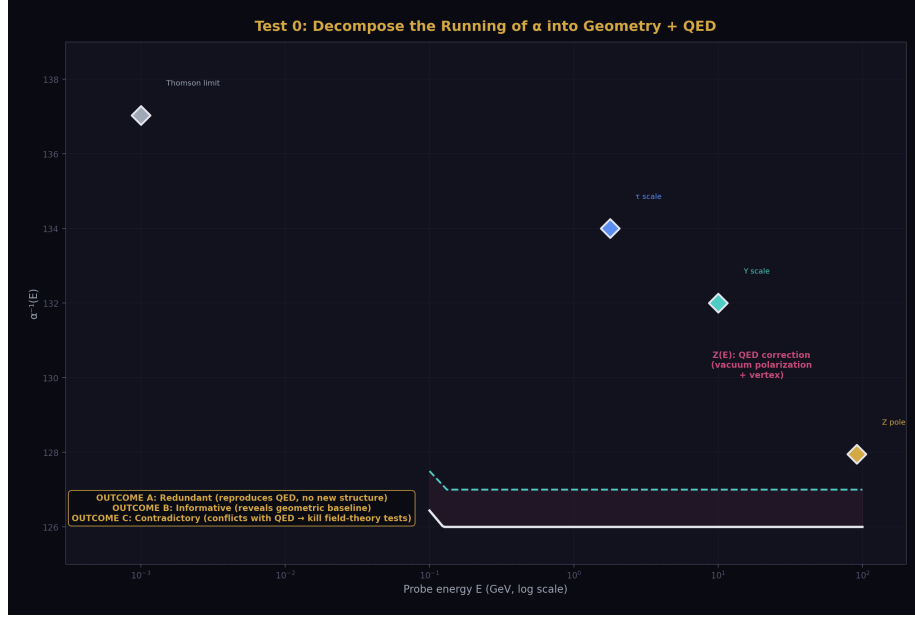


Figure 2: Fig. 5: The running of  $\alpha$  separated into geometric baseline  $\beta \cdot d^2(E)$  and QED correction  $Z(E)$  — three outcomes determine whether the framework extends to field-theory boundaries.

### 4.1 Why Calibrate

The framework must reproduce something known before it is applied to something unknown. The running of  $\alpha$  is the calibration case. The electron's vacuum polarization boundary is geometric (Section III.5). The interior and exterior readings are measured. The QED corrections that produce the running are published and precise. If  $Q = F \cdot \beta \cdot d^2 \cdot Z$  cannot engage with the known running of  $\alpha$  in a structurally meaningful way, it has no business being applied to unknown cases.

### 4.2 The Decomposition

A measurement of  $\alpha$  at probe energy  $E$  involves a scattering cross-section. The institution publishes these cross-sections — they are the primary observables of collider experiments. The total  $e^+e^- \rightarrow e^+e^-$  cross-section at each energy contains  $\pi r^2$  terms naturally. These are circular geometric cross-sections in the center-of-mass frame — exactly the type of quantity MATH-1 addresses.

Under the MATH-1 structure:

- $Q$  is the measured value of  $\alpha$  at energy  $E$

- $F$  is the electromagnetic driving term — the probe’s energy and charge configuration
- $\beta \cdot d^2(E)$  is the scattering cross-section at energy  $E$  — the published, measured circular cross-section
- $Z$  is the QED correction — vacuum polarization loops, vertex corrections, the domain-specific impedance

The critical step: rather than computing a geometric scale from first principles and risking an incorrect identification, the decomposition uses the published scattering cross-sections directly as  $\beta \cdot d^2$ . The cross-sections are measured. They are circular. They contain  $\beta \cdot d^2$  by construction — this is exactly MATH-1’s point. The  $Z$  extracted from the decomposition is whatever remains after the geometric cross-section is factored out.

#### 4.3 What to Look For

Three possible outcomes:

**Outcome A — Redundant.** The decomposition reproduces the known running exactly, with  $\beta \cdot d^2$  tracking the measured scattering cross-section and  $Z$  tracking the quantum corrections. The framework adds no new information. The running is fully captured by QED. The framework is consistent but tells us nothing we didn’t know.

This outcome does not kill the program. It confirms consistency with known physics. It provides no evidence that the framework adds value for known cases. Proceed to Tests 1-2 with reduced expectations.

**Outcome B — Structurally informative.** The decomposition separates the running into a geometric component ( $\beta \cdot d^2$ ) that varies smoothly with probe energy and a QED component ( $Z$ ) that captures the quantum corrections. The separation reveals a geometric baseline — a smooth function of scattering cross-section — from which the quantum corrections deviate. This baseline is implicit in QED but not separated or named. The decomposition makes it explicit.

This outcome supports the program. It shows the framework adds organizational value even for known physics. Proceed with confidence.

**Outcome C — Contradictory.** The decomposition produces inconsistencies —  $\beta \cdot d^2$  does not track the measured cross-section, or the  $Z$  residual does not match QED calculations, or the separation introduces artifacts. The framework contradicts known physics.

This outcome kills the program for field-theory boundaries. If the framework cannot handle the simplest geometric boundary in quantum field theory, it cannot be applied to more complex quantum boundaries. Tests 3, 5, and 6 are not motivated. Tests 1, 2, and 4 — which involve gravitational boundaries, not field-theory boundaries — may still be valid.



#### 4.4 The Calculation

Take published values of  $\alpha$  at different probe energies from LEP, SLD, and other collider measurements. For each energy, take the published scattering cross-section  $\sigma(E)$ . Identify  $\beta \cdot d^2(E)$  from  $\sigma(E)$ . Extract  $Z(E) = Q / (F \cdot \beta \cdot d^2(E))$ . Compare  $Z(E)$  to published QED vacuum polarization corrections at each energy.

This calculation uses published data and published QED results. It requires no new measurements. It is achievable now.

### V. TEST 1: G DECOMPOSITION BY METHOD

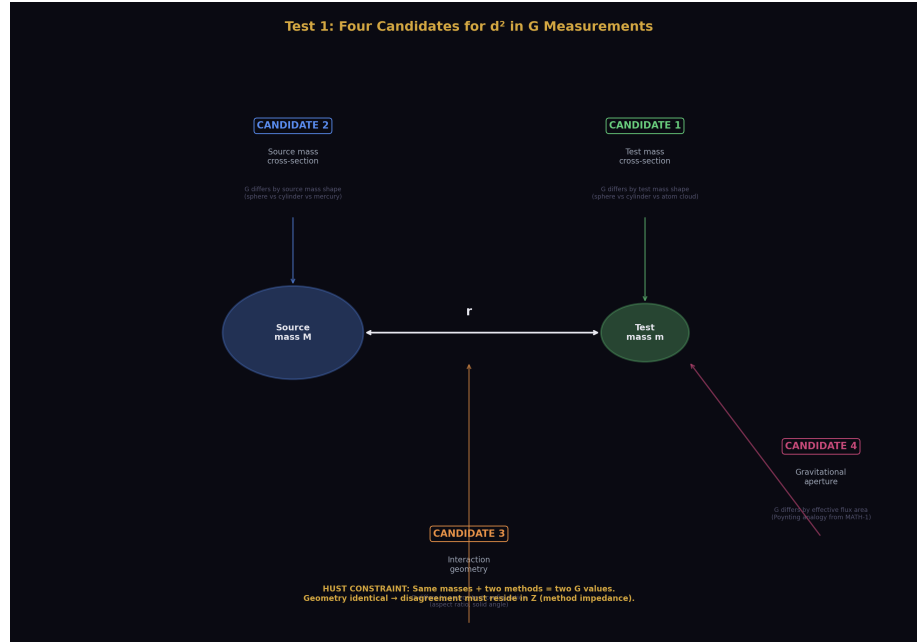


Figure 3: Fig. 6: Four candidate  $d^2$  identifications on a Cavendish apparatus — test mass, source mass, interaction geometry, gravitational aperture — constrained by the HUST internal control.

#### 5.1 The Question

[[HOWL-PHYS-3-2026](#)] documents persistent disagreement between G measurements from different experimental groups using different methods. If the  $Q = F \cdot \beta \cdot d^2 \cdot Z$  framework applies to G measurements, the disagreement should decompose into an invariant geometric component ( $\beta \cdot d^2$ ) and a variant apparatus-specific component ( $Z$ ).

## 5.2 The Decomposition

A Cavendish-type  $G$  measurement uses two masses (source mass  $M$ , test mass  $m$ ) separated by distance  $r$ . The measured quantity is the gravitational force  $F_{\text{grav}} = GMm/r^2$  or the corresponding acceleration or angular deflection.

Under the MATH-1 structure:

- $Q$  is the measured gravitational effect
- $F$  is the gravitational driving term — the source mass and field geometry
- $d^2$  is the bounding area of the relevant geometric cross-section
- $Z$  is the apparatus-specific impedance — torsion fiber properties, pendulum characteristics, atom interferometer configuration, vibration isolation, thermal environment
- $\beta \cdot d^2$  is the geometric invariant — the same across all methods if the framework applies

## 5.3 Two Candidates for $d^2$

**Candidate 1: Test mass cross-section.** The test mass has a physical cross-section. Different experiments use test masses of different sizes, shapes, and materials — spheres, cylinders, cold atom clouds. If the gravitational interaction is mediated through the geometric cross-section of the test mass, then  $d^2$  is the bounding area of the test mass.

Prediction: the  $G$  disagreement should correlate with test mass geometry. Spherical test masses should give systematically different values than cylindrical test masses of the same total mass.

**Candidate 2: Interaction geometry.** The gravitational field at distance  $r$  subtends a solid angle determined by the source mass geometry. The relevant cross-section is the area of the gravitational interaction region at the measurement distance. This is more abstract but would produce a single  $\beta \cdot d^2$  for a given source-test mass configuration regardless of individual mass shapes.

Prediction: the  $G$  disagreement should correlate with the source-test mass geometric configuration rather than with test mass shape alone.

## 5.4 An Internal Control

The HUST 2018 experiment produced two  $G$  values from the same laboratory, same location, same masses, using two different methods — angular acceleration feedback and time-of-swing. The values ( $6.67484$  and  $6.67349 \times 10^{-11}$ ) disagree. This pair is a natural internal control. The test mass geometry is identical for both methods. If Candidate 1 (test mass cross-section) fully explains the disagreement, this pair should produce identical results. It does not. Either Candidate 1 is incomplete, or the method-specific component ( $Z$ ) differs between the two techniques, or both geometric and method-specific factors contribute.

The HUST pair constrains the candidates. Any viable  $d^2$  identification must allow the HUST pair to disagree through  $Z$  while maintaining geometric invariance.

### 5.5 The Test

Reanalyze published  $G$  measurements. For each measurement, record test mass geometry, source mass geometry, source-test mass separation, orientation, and measured  $G$  value. Sort by geometric variables. Test for correlation.

If correlation with geometry exists: the framework applies to  $G$ . The geometric variable that correlates identifies the correct  $d^2$  candidate.

If no correlation with geometry exists: the framework does not explain the  $G$  disagreement through geometric cross-sections. The disagreement has a different source.

This test uses published data. It requires no new measurements. It is achievable now.

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## VI. TEST 2: $G$ DEPTH TREND

### 6.1 The Question

[[HOWL-PHYS-3-2026](#)] proposes that the  $G$  disagreement may reflect depth-dependent readings within Earth's Hill sphere. If so, measurements at different effective depths within Earth's gravitational structure should show a systematic trend.

### 6.2 The Test

Reanalyze published  $G$  measurements sorted by effective gravitational potential. For each measurement, record altitude above sea level, latitude, local gravitational acceleration, and the local gravitational potential. Plot  $G$  versus each variable. Test for systematic trend using non-parametric methods (Spearman rank correlation or equivalent).

If systematic trend exists: the depth-dependent reading interpretation is supported. The direction and magnitude constrain the boundary effect.

If random scatter with no trend: the depth interpretation is not supported.

### 6.3 Sensitivity Assessment

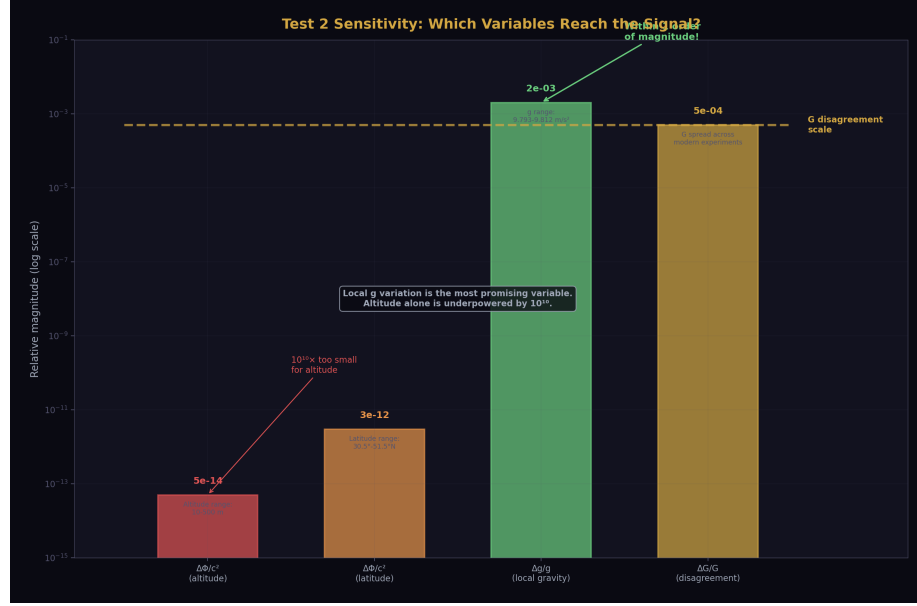


Figure 4: Fig. 7: Altitude variation is  $10^{10} \times$  too small. Local g variation is within one order of magnitude of the G disagreement — the most promising variable for Test 2.

The altitude range across modern precision G experiments is approximately 10 meters to 500 meters above sea level. The corresponding gravitational potential difference is  $\Delta\Phi/c^2 \sim 5 \times 10^{-14}$ . The G disagreement between methods is at the  $\sim 10^{-4}$  level. If the depth effect scales linearly with gravitational potential, the expected signal from altitude variation alone is many orders of magnitude below the observed disagreement. The test is likely underpowered for altitude.

Latitude provides a larger lever arm. The latitude range across modern experiments spans approximately  $30^\circ$ N (Wuhan) to  $52^\circ$ N (Wuppertal), producing local g variations of approximately 0.3%. This is still small relative to the G disagreement, but larger than the altitude variation.

The test is specified despite the sensitivity concern for three reasons. First, if a positive result appears at this precision, it would indicate a boundary effect far larger than gravitational potential scaling predicts — which would itself be a significant and constraining finding. Second, a null result establishes an upper bound on the depth effect, replacing the current state of no bound at all. Third, the test costs nothing — it is a reanalysis of existing data.

This test uses published data and published geodetic information. It requires no new measurements. It is achievable now.

## VII. TEST 3: PROTON RADIUS RESIDUAL



Figure 5: Fig. 2: Electron probes at Bohr radius (large cross-section, reads 0.877 fm) vs muon at  $a_0/207$  ( $42,849 \times$  smaller cross-section, reads 0.842 fm).

### 7.1 The Question

The proton radius puzzle: electron spectroscopy historically yields  $\sim 0.877$  fm, muonic spectroscopy yields  $\sim 0.842$  fm. PHYS-1 identifies this as a boundary-depth effect — different probes at different interaction depths read different radii. The proton boundary is geometric (Section III.4). Does the residual after QED corrections correlate with the probe cross-section ratio?

### 7.2 The Decomposition

The electron probe interacts at the Bohr radius  $a_0 \approx 5.29 \times 10^{-11}$  m. The interaction cross-section is circular:  $\beta \cdot d^2_e$  where  $d_e = 2a_0$ . The muon probe interacts at  $a_0/207 \approx 2.56 \times 10^{-13}$  m. The interaction cross-section is circular:  $\beta \cdot d^2_\mu$  where  $d_\mu = 2a_0/207$ .

Under the MATH-1 structure:

- $Q_e$  and  $Q_\mu$  are the measured proton radii from each probe
- $F_e$  and  $F_\mu$  are the electromagnetic driving terms
- $\beta \cdot d^2_e$  and  $\beta \cdot d^2_\mu$  are the geometric cross-sections at each interaction depth
- $Z_e$  and  $Z_\mu$  are the QED corrections specific to each probe

The cross-section ratio:  $d^2_e / d^2_\mu = 207^2 = 42,849$ .

### 7.3 Directional Prediction

The framework makes a directional prediction, not just a correlation test. The electron probe has a larger interaction cross-section (42,849 times larger). The electron measurement yields a larger proton radius (0.877 fm). The muon probe has a smaller cross-section. The muon measurement yields a smaller radius (0.842 fm). Larger cross-section corresponds to larger apparent radius. The direction of the puzzle is consistent with the framework's geometric structure — the probe with the larger geometric footprint reads a larger boundary.

This directional consistency is stated as a pre-existing feature of the data, not as a prediction made before seeing the data. The test of the framework is whether the magnitude of the discrepancy — not just its direction — correlates with the cross-section ratio through  $\beta$ .

### 7.4 The Test

Take published proton radius measurements and published QED corrections for both electron and muon probes. Compute the residual for each — the difference between measured radius and QED-predicted radius. Test whether the ratio of residuals correlates with the ratio of probe cross-sections ( $207^2$ ) or with some function of probe cross-section that  $\beta$  mediates.

If the residual ratio correlates with the cross-section ratio: the proton radius puzzle has a geometric component that QED corrections do not capture.

If the residual ratio does not correlate: the geometric framework does not explain the proton radius puzzle. The anomaly has a different source.

### 7.5 Complication

Recent electron-probe measurements have trended toward the muonic value, narrowing the puzzle. If the puzzle fully resolves through improved QED calculations or improved electron spectroscopy, Test 3 becomes moot — the geometric framework is not needed for this anomaly. This is a legitimate resolution and would satisfy PHYS-1's falsification criterion F2. The test must use the most current data and the most current QED corrections.

This test uses published data. It requires careful numerical work with existing QED calculations. It is achievable now.

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## VIII. TEST 4: G AT L1/L2

### 8.1 The Question

[[HOWL-PHYS-3-2026](#)] proposes the first direct measurement of G outside Earth's Hill sphere. The geometric framework adds structure: the Hill sphere is a geometric boundary

with circular cross-section. If the framework applies, the boundary-crossing measurement has geometric content.

## 8.2 The Measurement

A self-contained G measurement apparatus at L1 or L2 — two known masses, precision separation measurement, force or acceleration measurement — operated in the gravitationally quiet Lagrange point environment. The measured G value is compared to the Earth-surface consensus after applying all standard GR corrections for the different gravitational potential.

## 8.3 What the Framework Adds

Without the MATH-1 framework, the L1/L2 measurement is a test of whether G differs outside Earth's Hill sphere. The expected magnitude of any difference is unconstrained — the test is pass/fail with no quantitative prediction.

With the MATH-1 framework, the measurement has geometric structure. The Hill sphere is a sphere of known radius. The measurement apparatus has a known geometry. Whether the ratio of apparatus cross-section to Hill sphere cross-section — mediated by  $\beta$  — predicts the magnitude of any G difference is a testable question. This is stated as a structural possibility, not a derived prediction. The connection between MATH-1's geometric ratio and a boundary-crossing G measurement has not been derived from first principles. The framework provides a candidate structure. The measurement provides the data. The comparison determines whether the candidate is correct.

## 8.4 Status

This test requires a new instrument at L1 or L2. The location is occupied by functioning spacecraft. The measurement concept exists in the institution's published literature, proposed for precision reasons. The boundary-crossing motivation has been provided by PHYS-3. The geometric structure has been provided by MATH-1. The measurement has not been performed.

Achievable with current technology. Not yet designed for this purpose.

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# IX. TEST 5: HUBBLE TENSION

## 9.1 The Question

The Hubble tension: CMB measurements yield  $H_0 = 67.4 \pm 0.5$  km/s/Mpc, local measurements yield  $H_0 = 73.0 \pm 1.0$  km/s/Mpc. The ratio is 0.923. PHYS-1 identifies the correlation with boundary transit count — CMB light crosses more coherent structures than local supernova light. Does the cumulative geometric effect across boundary transits produce a factor consistent with 0.923?

## 9.2 What the Calculation Would Require

1. An estimate of the number of coherent gravitational structures between the surface of last scattering and our instruments
2. The characteristic cross-section and bounding area of each structure type
3. The per-boundary transit correction magnitude
4. The cumulative product across all transits
5. Comparison to the observed ratio of 0.923

The first two inputs are available from published large-scale structure surveys — the Sloan Digital Sky Survey, the 2dF Galaxy Redshift Survey, and others catalog the distribution of structures along the line of sight. The characteristic sizes of galaxy clusters, filaments, and voids are published.

## 9.3 The Missing Piece

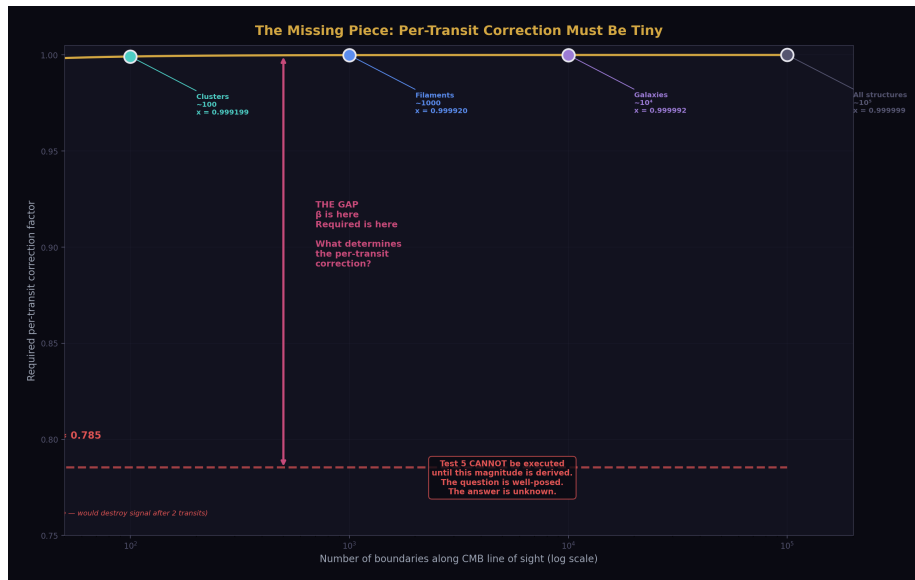


Figure 6: Fig. 4:  $\beta=0.785$  would destroy the signal after two transits. The required per-transit correction is  $\sim 0.9999$  for 100 boundaries — the gap between  $\beta$  and the required value is the missing theoretical piece.

The third input — the per-boundary transit correction magnitude — is not provided by any paper in this series. This is the primary limitation of Test 5.

MATH-1 establishes  $\beta = \pi / 4$  as the geometric ratio for cross-sectional area. But the transit correction for light crossing a gravitational boundary is not an area ratio. It is a



transformation of the light’s measured properties (frequency, flux, timing) as it crosses the boundary. MATH-1 does not derive this transformation. PHYS-1 identifies the boundary transit as an unmodeled element but does not quantify it.

The magnitude constraint is severe. If the per-transit correction were  $\beta$  itself (0.785), even two boundary transits would reduce the measured quantity by a factor of 0.616 — already far below the observed 0.923 ratio. Any line of sight to the CMB crosses hundreds to thousands of coherent structures. The per-transit correction must be extraordinarily small — many orders of magnitude smaller than  $\beta$  — or the cumulative effect would overwhelm the observed tension rather than explaining it.

Determining the per-boundary transit correction is the missing theoretical piece. It may be derivable from the boundary geometry, from the ratio of light wavelength to boundary size, or from some other structural relationship. This derivation has not been performed and is beyond the scope of this paper.

#### 9.4 Status

Test 5 is underspecified. The question is well-posed. The observational inputs are available. The per-transit correction magnitude is unknown. Without it, the calculation cannot be performed. Test 5’s viability depends on prior theoretical work that this paper identifies as needed but does not attempt.

Test 5 is reclassified from “partially achievable” to “requires prior theoretical work.” It remains in the program as a target — if the per-transit correction is derived (by this research program or by others), the test becomes immediately calculable from existing data.

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## X. TEST 6: GALACTIC BOUNDARY AND ROTATION CURVES

### 10.1 The Question

Galaxy rotation curves are flat at large radii where visible matter is insufficient to account for orbital velocities. The institution interprets this as dark matter. PHYS-1 proposes that the gravitational evidence requires inertia, not necessarily substance. The galactic boundary is geometric — approximately elliptical (Section III.7). Does the  $\beta$  framework apply to the galactic boundary in a way that affects the inferred mass distribution?

### 10.2 What the Test Requires

A galaxy’s gravitational boundary is approximately an oblate ellipsoid. MATH-1 establishes that  $\beta = \pi/4$  is the ratio of elliptical area to bounding rectangle at all eccentricities. The rotation curve depends on the enclosed mass as a function of radius. If the geometric framework applies, the enclosed mass calculation at each radius involves a  $\beta$ -corrected cross-section that differs from the standard calculation.

The question is whether the geometric correction modifies the inferred mass profile in a way that reduces the need for dark matter.

### **10.3 Status**

This is the most computationally difficult test in the program. The galactic boundary is not a clean ellipse. The rectilinear measurement frame is not obvious at galactic scales. The mass distribution is three-dimensional and complex. Additionally, the same missing per-transit correction that limits Test 5 may affect this test — the rotation curve discrepancy involves measurements of stars at different depths within the galactic boundary, and the depth-dependent reading may require the same per-transit correction.

The test is stated for completeness and future work. Its viability depends on the results of earlier tests. If Tests 0-4 all produce null results, this test is not motivated.

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## **XI. TEST 7: SOLAR SYSTEM HILL SPHERE**

### **11.1 The Question**

If G is boundary-dependent at the Earth Hill sphere scale, it should also show a reading change at the solar system Hill sphere scale — approximately 1-2 light years from the Sun, at the inner edge of the Oort Cloud.

### **11.2 Status**

No instrument has been placed at or beyond the solar system's Hill sphere. Voyager 1 and 2 have crossed the heliopause — the boundary of the solar wind at ~120 AU — but the gravitational Hill sphere is at ~50,000-100,000 AU. These are different boundaries by three orders of magnitude. This test is not achievable with current technology. It is stated as a long-term falsification target.

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## XII. THE TEST PROGRAM

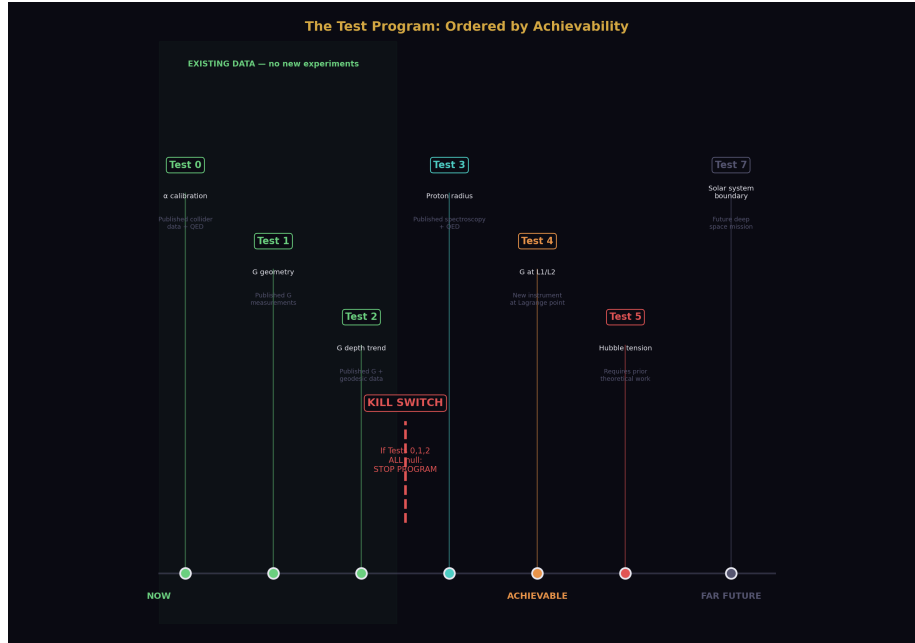


Figure 7: Fig. 3: Seven tests from existing-data reanalysis (now) through new instruments to far-future missions, with the kill switch gate after Tests 0-2.

### 12.1 Ordered by Achievability

| Priority | Test                            | Data Source                              | New Measurement? | Status                               |
|----------|---------------------------------|------------------------------------------|------------------|--------------------------------------|
| 0        | Running of $\alpha$ calibration | Published collider data + QED            | No               | Achievable now                       |
| 1        | G decomposition by method       | Published G measurements                 | No               | Achievable now                       |
| 2        | G depth trend                   | Published G measurements + geodetic data | No               | Achievable now (likely underpowered) |
| 3        | Proton radius residual          | Published spectroscopy + QED             | No               | Achievable now                       |
| 4        | G at L1/L2                      | New instrument at Lagrange point         | Yes              | Achievable, not yet done             |

| Priority | Test                  | Data Source                   | New Measurement? | Status                          |
|----------|-----------------------|-------------------------------|------------------|---------------------------------|
| 5        | Hubble tension        | Published structure surveys   | No (calculation) | Requires prior theoretical work |
| 6        | Rotation curves       | Published rotation curve data | No (calculation) | Difficult, requires modeling    |
| 7        | Solar system boundary | Future deep space mission     | Yes              | Not achievable now              |

## 12.2 The Kill Switch

Tests 0, 1, and 2 are reanalyses of existing published data. They require no new experiments, no new instruments, no funding. They are achievable by any researcher with access to the published literature.

If all three produce null results — the framework does not reproduce the running of  $\alpha$  in a structurally meaningful way (Outcome A or C), the G disagreement shows no correlation with geometry, and the G disagreement shows no depth trend — the program is not supported. Tests 3-7 are not motivated. The geometric framework does not extend from cross-section equations to physical boundaries. MATH-1 remains valid within its proven scope. PHYS-1's anomaly correlations remain qualitative. PHYS-3's G gap remains factual but unconnected to the geometric framework.

This is a genuine commitment. If the easy tests fail, the hard tests are not worth pursuing. The scope boundary — where the geometric framework stops — is itself a finding.

If any of Tests 0-2 produce positive results, the program is supported for further testing. The specific positive result determines which subsequent tests are most motivated.

## 12.3 Decision Tree

Test 0 ( $\alpha$  calibration) has three outcomes. Contradictory kills the program for field-theory boundaries but not for gravitational boundaries — Tests 1, 2, and 4 remain valid. Redundant means the framework is consistent but uninformative — proceed with reduced expectations. Informative means the framework adds structural value — proceed with confidence.

Tests 1 and 2 (G geometry and depth) each have two outcomes. The four combinations determine what follows:

Both null: program not supported, stop.

Test 1 positive only: geometric component in G disagreement, proceed to Test 4.

Test 2 positive only: depth-dependent readings, proceed to Test 4.

Both positive: strong support, proceed to Tests 3 and 4.

Test 0 contradictory combined with Tests 1 or 2 positive: the framework may apply to gravitational boundaries but not field-theory boundaries. Proceed to Test 4 only. Skip Tests 3, 5, 6.

After Tests 3-4, a second evaluation: if both null, the framework does not produce measurable effects at accessible boundaries. If any positive, proceed to Tests 5-6, using the positive results to constrain the per-transit correction magnitude needed for Test 5.

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### XIII. FALSIFICATION CRITERIA

**F1 — Calibration.** If the  $Q = F \cdot \beta \cdot d^2 \cdot Z$  decomposition of the running of  $\alpha$  produces results that contradict published QED calculations, the framework is inconsistent with established physics and cannot be applied to any quantum field theory boundary.

**F2 — G geometry.** If reanalysis of published G measurements sorted by test mass geometry, source mass geometry, and source-test mass configuration shows no correlation between G value and any geometric parameter, the geometric framework does not explain the G disagreement.

**F3 — G depth.** If reanalysis of published G measurements sorted by altitude, latitude, and local gravitational potential shows no systematic trend, the depth-dependent reading interpretation is not supported. A null result at the available precision establishes an upper bound on the depth effect but does not rule out effects below that bound.

**F4 — Proton radius.** If the residual of the proton radius puzzle after published QED corrections does not correlate with the ratio of probe interaction cross-sections, the geometric framework does not explain the proton radius puzzle. If the puzzle resolves through improved QED or spectroscopy without geometric correction, the framework is not needed for this anomaly.

**F5 — G at L1/L2.** If a direct measurement of G at L1 or L2 produces a value consistent with the Earth-surface consensus within measurement uncertainty after all GR corrections, the boundary effect hypothesis for G at the Earth Hill sphere scale is not supported at that precision level.

**F6 — Hubble tension.** If the per-transit correction is derived and the cumulative geometric transit effect does not produce a factor consistent with the observed  $H_0$  ratio, the geometric framework does not explain the Hubble tension.

**F7 — Overall.** If Tests 0, 1, and 2 all produce null results, the geometric framework does not extend from cross-section equations to physical boundaries. The program is not supported. The scope boundary is documented as a finding.

Each criterion is specific and stated before the evidence is examined.

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## XIV. BOUNDARIES AND LIMITATIONS

| PHYS-4 Identity Card                                                                                                                                                                                                   |                      |            |                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------|--------------------------------------------------------------------------|
| THE BOUNDARY TEST PROGRAM                                                                                                                                                                                              |                      |            |                                                                          |
| SEVEN TESTS                                                                                                                                                                                                            |                      |            | KEY PRINCIPLES                                                           |
| Test 0                                                                                                                                                                                                                 | $\alpha$ calibration | NOW        | <b>CALIBRATE FIRST</b><br>Reproduce known results before testing unknown |
| Test 1                                                                                                                                                                                                                 | G geometry           | NOW        |                                                                          |
| Test 2                                                                                                                                                                                                                 | G depth trend        | NOW        | <b>KILL SWITCH</b><br>Tests 0,1,2 all null $\Rightarrow$ program stops   |
| Test 3                                                                                                                                                                                                                 | Proton radius        | NOW        |                                                                          |
| Test 4                                                                                                                                                                                                                 | G at L1,L2           | NEW INST.  | <b>SCOPE BOUNDARY</b><br>6 geometric boundaries 1 excluded (confinement) |
| Test 5                                                                                                                                                                                                                 | Hubble tension       | THEORY     |                                                                          |
| Test 7                                                                                                                                                                                                                 | Solar boundary       | FAR FUTURE | <b>MISSING PIECE</b><br>Per-transit correction magnitude unknown         |
| <b>THE QUESTION</b><br>Does $Q = F \cdot \beta \cdot d^2 \cdot Z$ ,<br>proven for cross-sections across nine domains,<br>apply to the coherent pattern boundaries<br>where measurement anomalies have been documented? |                      |            |                                                                          |
| HOWL-PHYS-4: Seven tests. Three achievable now. Kill switch at the gate. Calibrate before applying.                                                                                                                    |                      |            |                                                                          |

Figure 8: Fig. 8: Seven tests ordered by achievability, kill switch at the gate, six geometric boundaries, one missing piece — calibrate before applying.

This paper is a test program, not a result. It specifies tests. It does not claim outcomes.

The framework may not apply to non-geometric boundaries. Section III classifies each boundary and excludes hadron confinement as outside scope.

The quantitative predictions for Tests 3-6 require calculations this paper does not perform. The paper specifies the calculations. The calculations are future work.

Test 5 is underspecified. The per-boundary transit correction magnitude is unknown. Without it, the Hubble tension calculation cannot be performed. This is the most significant limitation of the program as currently defined.

Test 2 is likely underpowered. The altitude range across existing G experiments corresponds to a gravitational potential difference far below the scale of the G disagreement. A null result constrains but does not eliminate the depth interpretation. A positive result at this precision would be surprising and highly constraining.

The connection between MATH-1's geometric invariant and PHYS-1's physical boundaries has not been derived from first principles. It is proposed as a structural hypothesis. The tests determine whether the hypothesis is productive.

The kill switch is genuine. If the easy tests fail, the program stops.

## APPENDIX A: BOUNDARY CLASSIFICATION — COMPLETE

| Boundary                  | Source Paper | Physical Structure                                                | Geometry Type                         | Interior Reading                               | Exterior Reading                           | Difference     | Reading MATH-1 Applies?                            | Test Case            | Test Priority      |
|---------------------------|--------------|-------------------------------------------------------------------|---------------------------------------|------------------------------------------------|--------------------------------------------|----------------|----------------------------------------------------|----------------------|--------------------|
| Electron vacuum cloud     | PHYS-1       | Virtual particle cloud, spherically symmetric                     | Sphere / scattering cross-section     | $\alpha \approx 1/127$ (bare charge at high E) | $\alpha \approx 1/137$ (screened at low E) | $\sim 8\%$     | Yes<br>—<br>scattering cross-sections are circular | Running of $\alpha$  | 0 (calibration)    |
| Earth Hill sphere         | PHYS-3       | Gravitational sphere of influence, $r \approx 1.5 \times 10^6$ km | Sphere                                | Earth gravity dominates                        | Solar field dominates                      | Untested for G | Yes<br>—<br>sphere with circular cross-section     | G measurement        | 1, 2, 4            |
| Proton                    | PHYS-1       | Composite quark structure, charge radius $\sim 0.84$ fm           | Measurement cross-section is circular | $0.842$ fm (muon probe, deeper)                | $0.877$ fm (electron probe, shallower)     | $\sim 4\%$     | Yes<br>—<br>scattering cross-sections are circular | Proton radius puzzle | 3                  |
| Observable universe / CMB | PHYS-1       | Surface of last scattering                                        | Sphere                                | $H_0 = 67.4$ (CMB, all boundaries transited)   | $H_0 = 73.0$ (local, fewer boundaries)     | $\sim 9\%$     | Yes<br>—<br>spherical boundary                     | Hubble tension       | 5 (underspecified) |

| Boundary                               | Source<br>Paper | Physical<br>Struc-<br>ture                                                                   | Geometry<br>Type           | Interior<br>Read-<br>ing                     | Exterior<br>Read-<br>ing                        | Reading<br>Dif-<br>fer-<br>ence        | MATH-<br>1<br>Ap-<br>plies?                       | Test<br>Case               | Test<br>Prior-<br>ity  |
|----------------------------------------|-----------------|----------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------|-------------------------------------------------|----------------------------------------|---------------------------------------------------|----------------------------|------------------------|
| Galaxy                                 | PHYS-<br>1      | Disk<br>+ halo<br>gravi-<br>ta-<br>tional<br>struc-<br>ture                                  | Oblate<br>ellip-<br>soid   | Flat<br>rota-<br>tion<br>curves              | Keplerian<br>de-<br>cline<br>ex-<br>pected      | Dark<br>mat-<br>ter<br>at-<br>tributed | Yes<br>— el-<br>lipse<br>gener-<br>aliza-<br>tion | Rotation<br>curves         | 6                      |
| Solar<br>sys-<br>tem<br>Hill<br>sphere | PHYS-<br>3      | Sun's<br>gravi-<br>ta-<br>tional<br>sphere<br>of<br>influ-<br>ence,<br>$r \approx$<br>1-2 ly | Sphere                     | Solar<br>dy-<br>nam-<br>ics<br>domi-<br>nate | Galactic<br>field<br>domi-<br>nates             | Untested                               | Yes<br>—<br>sphere                                | G<br>mea-<br>sure-<br>ment | 7 (far<br>fu-<br>ture) |
| Hadron<br>con-<br>fine-<br>ment        | PHYS-<br>1      | QCD<br>vac-<br>uum<br>struc-<br>ture                                                         | Momentum<br>space<br>scale | weak<br>(asymptotic<br>free-dom)             | $\alpha_s$<br>strong<br>(con-<br>fine-<br>ment) | Order<br>of<br>mag-<br>ni-<br>tude     | No —<br>not<br>spa-<br>tial<br>geom-<br>etry      | Outside<br>scope           | —                      |

## APPENDIX B: DATA REQUIREMENTS BY TEST



| Test                     | Published Data Needed                                                                                                     | Specific Sources                                                                                                                                                             | Calculation Required                                                                                                                            | Output Format                                                                                                       | Status         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------|
| 0 — $\alpha$ calibration | $\alpha$ values at multiple probe energies; scattering cross-sections at each energy; QED vacuum polarization corrections | LEP electroweak working group; SLD; PDG Review of Particle Physics; published $\sigma(e^+e^-)$ at each energy                                                                | Identify $\beta \cdot d^2$ from published $\sigma(E)$ ; extract $Z(E) = Q/(F \cdot \beta \cdot d^2)$ ; compare $Z$ to published QED corrections | Table: energy, $\alpha$ measured, $\sigma(E)$ , $\beta \cdot d^2$ extracted, $Z$ extracted, $Z$ published, residual | Achievable now |
| 1 — $G$ geometry         | $G$ values from all modern precision experiments; complete apparatus descriptions including mass geometries               | CODATA $G$ task group; Gundlach & Merkowitz 2000; Quinn et al. 2001, 2014; Rosi et al. 2014; Luo et al. 2009; Xue et al. 2018; original publications with apparatus diagrams | Record test mass shape, source mass shape, separation, orientation; sort by geometric parameters; compute correlation                           | Scatter plots: $G$ vs each geometric variable; correlation coefficients; significance levels                        | Achievable now |

| Test              | Published Data Needed                                                                     | Specific Sources                                                                                                                                                        | Calculation Required                                                                                                                        | Output Format                                                                                     | Status                  |
|-------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------|
| 2 — G depth       | G values from all modern experiments; location metadata                                   | Same G sources as Test 1; USGS geodetic data; published laboratory coordinates and elevations                                                                           | Record altitude, latitude, local g, gravitational potential; sort by each; test for monotonic trend                                         | Plot: G vs gravitational potential; Spearman $\rho$ and significance; upper bound on depth effect | Achievable now          |
| 3 — Proton radius | Electron and muon proton radius measurements; QED bound-state corrections for both probes | Pohl et al. 2010; Bezginov et al. 2019; Xiong et al. 2019; Fleurbaey et al. 2018; CODATA proton radius; published QED correction terms for hydrogen and muonic hydrogen | Compute residual for each probe (measured minus QED predicted); compute ratio of residuals; compare to cross-section ratio 207 <sup>2</sup> | Ratio: residual $\mu$ /residual e vs 207 <sup>2</sup> ; directional check; significance           | Achievable now          |
| 4 — G at L1/L2    | New measurement — no existing data                                                        | Future instrument                                                                                                                                                       | Direct G measurement at Lagrange point; apply GR corrections; compare to CODATA surface value                                               | G(L1/L2) $\pm$ uncertainty vs G(surface) $\pm$ uncertainty; residual after GR                     | Requires new instrument |

| Test                | Published Data Needed                                               | Specific Sources                                                                      | Calculation Required                                                                                                | Output Format                                                  | Status                                                           |
|---------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| 5 — Hubble tension  | Large-scale structure catalogs; cluster/filament/void distributions | SDSS; 2dF; Planck lensing maps; published void catalogs (Pan et al., Nadathur et al.) | Estimate boundary count and sizes along CMB line of sight; apply per-transit correction; compute cumulative product | Cumulative product vs 0.923                                    | Requires prior theoretical work (per-transit correction unknown) |
| 6 — Rotation curves | Published galaxy rotation curves; visible mass distributions        | de Blok et al. 2008 (THINGS); Sofue & Rubin 2001; published mass models               | Apply $\beta$ correction at galactic elliptical boundary; recompute expected rotation curve                         | Observed vs $\beta$ -corrected predicted vs standard predicted | Difficult — requires modeling                                    |
| 7 — Solar system    | Future deep space mission data                                      | Future mission                                                                        | Direct G measurement at ~1-2 light years                                                                            | G(boundary) vs G(interior)                                     | Not achievable now                                               |

## APPENDIX C: EXPECTED OUTCOMES AND IMPLICATIONS

| Test                     | Positive Result                                         | Positive Implies                                                            | Null Result                                        | Null Implies                                                  | Contradictory Result          | Contradictory Implies                                                        |
|--------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| 0 — $\alpha$ calibration | Decomposition reveals geometric baseline in QED running | Framework adds structural insight to known physics; calibration passed      | Decomposition reproduces QED without new structure | Framework consistent but redundant; proceed cautiously        | Decomposition contradicts QED | Framework wrong for field-theory boundaries; do not proceed to Tests 3, 5, 6 |
| 1 — G geometry           | G correlates with geometric parameters                  | G disagreement has geometric component; $d^2$ candidate identified          | No correlation with geometry                       | Disagreement — not geometric                                  | —                             | —                                                                            |
| 2 — G depth              | Monotonic trend with altitude or potential              | Depth-dependent readings within Earth boundary                              | Random scatter                                     | Disagreement — not depth-dependent; upper bound established   | —                             | —                                                                            |
| 3 — Proton radius        | Residual ratio tracks cross-section ratio               | Proton puzzle has unmodeled geometric component                             | Residual ratio does not track                      | Puzzle has non-geometric source (or is resolved by QED)       | —                             | —                                                                            |
| 4 — G at L1/L2           | G differs from surface value beyond GR corrections      | First boundary-crossing G measurement; boundary-dependent reading confirmed | G matches within uncertainty                       | No boundary effect at Earth Hill sphere at measured precision | —                             | —                                                                            |

| Test                | Positive Result                                    | Positive Implies                                      | Null Result          | Null Implies                                | Contradictory Result | Contradictory Implies |
|---------------------|----------------------------------------------------|-------------------------------------------------------|----------------------|---------------------------------------------|----------------------|-----------------------|
| 5 — Hubble tension  | Cumulative product $\approx 0.923$                 | Geometric transit explains $H_0$ discrepancy          | Product $\neq 0.923$ | Tension has non-geometric source            | —                    | —                     |
| 6 — Rotation curves | $\beta$ correction reduces dark matter requirement | Boundary geometry contributes to rotation discrepancy | Negligible effect    | Dark matter unaffected by boundary geometry | —                    | —                     |
| 7 — Solar system    | G differs at solar Hill sphere                     | G boundary-dependent at solar system scale            | G matches            | G consistent at solar system scale          | —                    | —                     |

## APPENDIX D: G MEASUREMENT RECORD WITH GEOMETRIC PARAMETERS

| Experiment             | Year | $G$ ( $\times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ ) | Method          | Test Mass Shape | Source Mass Shape | Separation (m) | Altitude (m) | Latitude              | Local g ( $\text{m/s}^2$ ) |
|------------------------|------|----------------------------------------------------------------------|-----------------|-----------------|-------------------|----------------|--------------|-----------------------|----------------------------|
| Cavendish              | 1798 | 6.74                                                                 | Torsion balance | Sphere          | Sphere            | $\sim 0.23$    | $\sim 10$    | $51.5^\circ \text{N}$ | $\sim 9.812$               |
| NIST (Luther & Towler) | 1982 | 6.6726                                                               | Torsion balance | Cylinder        | Cylinder          | $\sim 0.10$    | $\sim 100$   | $39.1^\circ \text{N}$ | $\sim 9.801$               |
| Wuppertal (Michaelis)  | 1995 | 6.7154                                                               | Torsion balance | Cylinder        | Cylinder          | $\sim 0.10$    | $\sim 200$   | $51.3^\circ \text{N}$ | $\sim 9.811$               |

| Experiment                 | Year | $G \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ | Method                        | Test Mass Shape | Source Mass Shape | Separation (m) | Altitude (m) | Latitude | Local g ( $\text{m/s}^2$ ) |
|----------------------------|------|----------------------------------------------------------------|-------------------------------|-----------------|-------------------|----------------|--------------|----------|----------------------------|
| UWash/Erin Wash (Gundlach) | 2000 | 6.67422                                                        | Rotating torsion balance      | Flat disk/ring  | Sphere array      | ~0.10          | ~50          | 47.7°N   | ~9.808                     |
| BIPM (Quinn)               | 2001 | 6.67559                                                        | Torsion balance               | Cylinder        | Cylinder          | ~0.12          | ~400         | 48.8°N   | ~9.809                     |
| Zurich (Schlamminger)      | 2006 | 6.67425                                                        | Beam balance                  | Mass on beam    | Mercury tanks     | ~0.30          | ~500         | 47.4°N   | ~9.807                     |
| HUST (Luo)                 | 2009 | 6.67349                                                        | Torsion pendulum              | Sphere          | Sphere            | ~0.08          | ~30          | 30.5°N   | ~9.793                     |
| LENS (Rosi)                | 2014 | 6.67191                                                        | Atom interferometry           | Cold atom cloud | Tungsten masses   | ~0.26          | ~50          | 43.8°N   | ~9.804                     |
| BIPM (Quinn)               | 2014 | 6.67545                                                        | Torsion balance               | Cylinder        | Cylinder          | ~0.12          | ~400         | 48.8°N   | ~9.809                     |
| HUST (Xue) method 1        | 2018 | 6.67484                                                        | Angular acceleration feedback | Sphere          | Sphere            | ~0.08          | ~30          | 30.5°N   | ~9.793                     |
| HUST (Xue) method 2        | 2018 | 6.67349                                                        | Time of swing                 | Sphere          | Sphere            | ~0.08          | ~30          | 30.5°N   | ~9.793                     |

**Notes for Test 1:** Sort by test mass shape (sphere vs cylinder vs flat disk vs atom cloud). Compute mean  $G$  for each category. Test for systematic difference. Sort by source mass shape. Sort by separation. Sort by source-test configuration (aspect ratio, relative sizes). Test each geometric variable independently.

**Notes for Test 2:** Sort by altitude (range: 10-500 m). Sort by latitude (range: 30.5°-51.5°N). Compute local gravitational potential  $\Phi = -GM/r$  for each site using published geodetic models. Plot  $G$  vs  $\Phi$ . Apply Spearman rank correlation. The potential range

$\Delta\Phi/c^2 \approx 5 \times 10^{-14}$  is far below G measurement precision — a positive result at this range would indicate an unexpectedly large boundary effect.

**HUST internal control:** Methods 1 and 2 use same masses, same location, same separation. Only the measurement method differs. G values disagree (6.67484 vs 6.67349). This constrains  $d^2$  candidates: the geometric component should be identical for both methods. The disagreement must reside entirely in Z (method-specific impedance). If the HUST pair shows the same discrepancy magnitude as the cross-method spread generally, the geometric component is not isolated by controlling mass geometry.

## APPENDIX E: PROTON RADIUS DATA

| Measurement                                   | Year | Probe    | Probe Mass (MeV/c <sup>2</sup> ) | Orbital Radius                                   | Cross-Section $\beta \cdot d^2$                                     | Measured Radius (fm) | Uncertainty (fm) |
|-----------------------------------------------|------|----------|----------------------------------|--------------------------------------------------|---------------------------------------------------------------------|----------------------|------------------|
| CODATA 2014 (hydrogen spectroscopy consensus) | 2014 | Electron | 0.511                            | $a_0 \approx 5.29 \times 10^{-11} \text{ m}$     | $\beta \cdot (2a_0)^2 \approx 8.80 \times 10^{-21} \text{ m}^2$     | 0.8751               | $\pm 0.0061$     |
| Pohl et al. (muonic hydrogen)                 | 2010 | Muon     | 105.7                            | $a_0/207 \approx 2.56 \times 10^{-13} \text{ m}$ | $\beta \cdot (2a_0/207)^2 \approx 2.05 \times 10^{-25} \text{ m}^2$ | 0.84184              | $\pm 0.00067$    |
| Bezginov et al. (hydrogen 2S-2P)              | 2019 | Electron | 0.511                            | $a_0$                                            | $\approx 8.80 \times 10^{-21} \text{ m}^2$                          | 0.833                | $\pm 0.010$      |
| Xiong et al. / PRad (e-p scattering)          | 2019 | Electron | 0.511                            | Varies with $Q^2$                                | Varies                                                              | 0.831                | $\pm 0.012$      |
| Fleurbay et al. (hydrogen 1S-3S)              | 2018 | Electron | 0.511                            | $a_0$                                            | $\approx 8.80 \times 10^{-21} \text{ m}^2$                          | 0.877                | $\pm 0.013$      |

**Cross-section ratio:**  $d^2_{\text{electron}} / d^2_{\text{muon}} = (2a_0)^2 / (2a_0/207)^2 = 207^2 = 42,849$

**Directional check:** Larger cross-section (electron) → larger measured radius (0.877 fm). Smaller cross-section (muon) → smaller measured radius (0.842 fm). Direction is consistent with geometric framework prediction: larger geometric footprint reads larger boundary.

**Convergence note:** Bezginov 2019 and Xiong/PRad 2019 electron measurements yield values near the muonic value, narrowing the puzzle. If convergence continues, Test 3 becomes moot — the puzzle resolves without geometric correction. Test 3 must use the most current data and most current QED corrections at the time of execution.

## APPENDIX F: RUNNING OF $\alpha$ — CALIBRATION DATA

| Energy Scale                     | $\alpha^{-1}$ Measured | Source                           | Scattering Cross-Section $\sigma(E)$                  | Published QED Correction   | Z to Extract           |
|----------------------------------|------------------------|----------------------------------|-------------------------------------------------------|----------------------------|------------------------|
| 0 (Thomson limit)                | 137.036                | Fundamental — static measurement | —                                                     | Fully screened             | Baseline               |
| ~1 GeV ( $\tau$ mass scale)      | ~134                   | $e^+e^-$ collider data           | Published $\sigma(e^+e^- \rightarrow \text{hadrons})$ | VP + vertex                | Compare Z to published |
| ~10 GeV ( $\Upsilon$ mass scale) | ~132                   | $e^+e^-$ collider data           | Published $\sigma(e^+e^-)$                            | VP + vertex + higher order | Compare Z to published |
| ~45 GeV (Z/2 scale)              | ~129                   | LEP I                            | Published $\sigma$ at $\sqrt{s} = 45$ GeV             | VP + EW corrections        | Compare Z to published |
| 91.2 GeV (Z mass)                | $127.95 \pm 0.016$     | LEP I / SLD                      | Published $\sigma$ at Z pole                          | Full EW + QCD corrections  | Compare Z to published |

### Test 0 protocol:

Step 1 — For each energy E, take the published scattering cross-section  $\sigma(E)$ . These are measured values from collider experiments — primary observables, not derived quantities.

Step 2 — Each  $\sigma(E)$  is a circular cross-section. It contains  $\beta \cdot d^2$  by construction. Identify  $\beta \cdot d^2(E)$  from  $\sigma(E)$  by factoring out the electromagnetic coupling and kinematic factors.

Step 3 — Express  $\alpha(E) = Q/(F \cdot \beta \cdot d^2(E))$ . Extract  $Z(E)$  as the residual.



Step 4 — Compare  $Z(E)$  to published QED vacuum polarization corrections at each energy. The published corrections include hadronic vacuum polarization, leptonic vacuum polarization, and vertex corrections.

Step 5 — Evaluate outcome: Does the  $\beta \cdot d^2$  component track a smooth geometric baseline? Does  $Z$  match published QED? Is there a residual?

**Key distinction from naive approach:** This protocol uses measured cross-sections, not computed geometric radii. The naive identification  $r(E) = \hbar c/E$  conflates the probe wavelength with the physical screening radius. The published cross-section is the correct geometric input because it is the measured circular cross-section at each energy — exactly the quantity MATH-1 addresses.

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## APPENDIX G: HUBBLE TENSION — STRUCTURAL DATA

| Method                              | $H_0$<br>(km/s/Mpc) | Uncertainty | Estimated<br>Major<br>Boundary<br>Transits                                               | Notes                                                                               |
|-------------------------------------|---------------------|-------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Planck CMB                          | 67.4                | $\pm 0.5$   | Maximum —<br>all structure<br>between<br>surface of last<br>scattering and<br>instrument | Light has<br>crossed every<br>large-scale<br>structure in<br>observable<br>universe |
| SH0ES (Type<br>Ia SNe)              | 73.0                | $\pm 1.0$   | Moderate —<br>local group<br>structures                                                  | Supernovae in<br>relatively<br>nearby<br>galaxies                                   |
| H0LiCOW<br>(lensing time<br>delays) | 73.3                | $\pm 1.8$   | Intermediate<br>— includes<br>lens galaxy                                                | Quasar light<br>through<br>intervening<br>galaxy                                    |
| CCHP<br>(TRGB)                      | 69.8                | $\pm 1.7$   | Moderate —<br>similar to<br>SH0ES                                                        | Different<br>calibration<br>chain                                                   |
| DES + BAO +<br>BBN                  | 67.4                | $\pm 1.2$   | High —<br>cosmological<br>scale                                                          | Model-<br>dependent,<br>similar to<br>CMB                                           |

**Observed ratio:**  $67.4 / 73.0 = 0.923$

**Per-transit magnitude problem:** Any line of sight to the CMB crosses hundreds to

thousands of coherent gravitational structures (galaxy clusters, filaments, walls, voids).  
Published estimates from large-scale structure surveys:

| Structure Type                           | Estimated Count<br>Along CMB Line<br>of Sight | Typical Diameter         | Published Source                        |
|------------------------------------------|-----------------------------------------------|--------------------------|-----------------------------------------|
| Galaxy clusters                          | ~100-200                                      | 2-5 Mpc                  | SDSS cluster<br>catalogs                |
| Filaments                                | ~1000+                                        | 5-10 Mpc width           | Bond et al. 1996;<br>Cautun et al. 2014 |
| Voids                                    | ~50-100 (major<br>voids >30 Mpc)              | 30-100 Mpc               | Pan et al. 2012;<br>Nadathur 2016       |
| Individual galaxies<br>(including halos) | ~10 <sup>4</sup> -10 <sup>5</sup>             | 0.1-1 Mpc (with<br>halo) | Galaxy luminosity<br>function surveys   |

**Constraint on per-transit correction:** If correction per transit =  $x$ , then  $x^N \approx 0.923$  where  $N$  is the effective number of boundaries. For  $N = 100$ :  $x \approx 0.9992$ . For  $N = 1000$ :  $x \approx 0.99992$ . For  $N = 10,000$ :  $x \approx 0.999992$ . The per-transit correction must be extraordinarily close to 1.  $\beta$  itself (0.785) is far too large — it would predict essentially zero received signal after even a few transits.

**Status:** Test 5 cannot be executed without the per-transit correction magnitude. The boundary count and structure size inputs are available from published surveys. The missing piece is theoretical: what determines the per-transit correction, and how does it relate to  $\beta$ , to the boundary geometry, or to the ratio of wavelength to boundary size?

## APPENDIX H: DECISION TREE — COMPLETE

START

```

|
|— Test 0: a calibration
|   |— CONTRADICTORY
|   |   → Framework fails for field-theory boundaries
|   |   → Tests 3, 5, 6 not motivated (all involve field/quantum boundaries)
|   |   → Tests 1, 2, 4 still valid (gravitational boundaries only)

```

```

|   |   → Proceed to Tests 1-2
|   |
|   └─ REDUNDANT
|   |   → Framework consistent with QED, no new structure revealed
|   |   → Calibration neither passed nor failed –
inconclusive
|   |   → Proceed to Tests 1-2 with reduced expectations
|   |
|   └─ INFORMATIVE
|       → Framework reveals geometric structure in QED running
|       → Calibration passed
|       → Proceed to Tests 1-2 with confidence
|
└─ Tests 1-2: G geometry and G depth (run in parallel)
|   |
|   └─ BOTH NULL
|   |   → G disagreement is neither geometric nor depth-
dependent
|   |   → EVALUATION GATE: combined with Test 0 result
|   |   |
|   |   └─ Test 0 also null/redundant → ALL THREE NULL → STOP PROGRAM
|   |   |   Framework does not extend to physical boundaries
|   |   |   Scope boundary is the finding
|   |   |   All prior papers unaffected within their scope

```

- | | |
- | | └─ Test 0 informative → One positive exists
- | |     → Framework adds value for QED but not for G
- | |     → Proceed to Test 3 only (proton is quantum boundary)
- | |     → Skip Test 4 (gravitational boundary tests null)
- | |
- | └─ TEST 1 POSITIVE ONLY
- | |     → G disagreement has geometric component
- | |     → Which  $d^2$  candidate correlates? → Informs Test 4 design
- | |     → Proceed to Test 4 (G at L1/L2) as next priority
- | |     → Test 3 also motivated if Test 0 was not contradictory
- | |
- | └─ TEST 2 POSITIVE ONLY
- | |     → Depth-dependent readings within Earth boundary
- | |     → Trend direction and magnitude constrain boundary effect
- | |     → Proceed to Test 4 (G at L1/L2) as next priority
- | |     → Test 3 also motivated if Test 0 was not contradictory
- | |
- | └─ BOTH POSITIVE
- |     → Strong support for geometric and depth structure in G
- |     → Proceed to Tests 3 and 4 in parallel
- |     → Tests 5-6 provisionally motivated pending Tests 3-4 results
- |

- |— SECOND STAGE: Tests 3-4
  - |
    - |— Test 3: Proton radius residual
      - | | — NULL → Proton puzzle not geometric; or puzzle resolved by QED
        - | | | — POSITIVE → Geometric component identified
          - | | | → Magnitude constrains boundary transformation law
          - | | | → Informs per-transit correction for Test 5
    - | | — Test 4: G at L1/L2
      - | | | — MATCHES → No boundary effect at Earth Hill sphere
        - | | | | → If Test 3 also null → Framework does not produce measurable effects
          - | | | | | → Stop further tests
      - | | | — DIFFERS → First cross-boundary G measurement in history
        - | | | → Direction and magnitude determine boundary law
        - | | | → Combined with any Test 3 result → proceed to Tests 5-6
  - | | — SECOND EVALUATION
    - | | | — BOTH NULL → Framework does not produce measurable effects
      - | | | | at accessible boundaries; Tests 5-7 not motivated
    - | | | — ANY POSITIVE → Proceed to Tests 5-6
      - | | | | Use positive results to constrain per-transit correction

```

|      └─ BOTH POSITIVE → Strong program; full continuation
|
|─ THIRD STAGE: Tests 5-6 (only if second stage supports)
|  |
|  |─ Test 5: Hubble tension (requires per-
transit correction first)
|  |─ INCONSISTENT → Geometric transit does not explain Hubble tension
|  |─ CONSISTENT → Cumulative boundary transit explains  $H_0$  discrepancy
|  |
|  |─ Test 6: Rotation curves
|  |─ NO EFFECT → Dark matter unaffected
|  |─ REDUCES REQUIREMENT → Geometric boundary contributes
|
|─ FOURTH STAGE: Test 7 (far future, only if program strongly supported)
|─ G MATCHES → Consistent at solar system scale
|─ G DIFFERS → Boundary-dependent at solar system scale

```

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## APPENDIX I: FALSIFICATION MATRIX

| Criterion | Test | What Is Tested                 | Falsifying Result                                    | Confidence Required                                | Current Status |
|-----------|------|--------------------------------|------------------------------------------------------|----------------------------------------------------|----------------|
| F1        | 0    | Framework consistency with QED | Decomposition contradicts published QED calculations | Any significant inconsistency with established QED | Not yet tested |

| Criterion | Test    | What Is Tested                             | Falsifying Result                                                                   | Confidence Required                                                                                | Current Status                      |
|-----------|---------|--------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------|
| F2        | 1       | G disagreement has geometric structure     | No correlation between G and any geometric parameter of apparatus                   | $p > 0.05$ across all geometric variables                                                          | Not yet tested                      |
| F3        | 2       | G is depth-dependent within Earth boundary | No trend with altitude, latitude, or gravitational potential                        | Spearman $\rho$ consistent with zero at available precision                                        | Not yet tested                      |
| F4        | 3       | Proton puzzle has geometric component      | Residual ratio does not track probe cross-section ratio; or puzzle resolves via QED | Ratio differs by $>2\sigma$ from geometric prediction; or electron values converge to muonic value | Not yet tested                      |
| F5        | 4       | G changes across Hill sphere boundary      | $G(L1/L2)$ equals $G(\text{surface})$ after GR corrections                          | Within measurement uncertainty of new instrument                                                   | Not yet tested                      |
| F6        | 5       | Cumulative transit explains Hubble tension | Cumulative product $\neq 0.923$ (once per-transit correction is known)              | Differs by $>10\%$ from observed ratio                                                             | Not yet tested; test underspecified |
| F7        | Overall | Framework extends to physical boundaries   | Tests 0, 1, and 2 all null                                                          | All three produce no positive signal                                                               | Not yet tested                      |

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## APPENDIX J: CONNECTION TO SERIES PAPERS

| Paper  | Registry                                     | Central Claim                                                                                                                       | Contributes to<br>PHYS-4                                                                                                    | PHYS-4 Adds                                                                                                                   |
|--------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| PHYS-1 | <a href="#">[HOWL-<br/>PHYS-1-<br/>2026]</a> | Mass is inertia;<br>particles are<br>field vortices;<br>three<br>anomalies<br>correlate with<br>boundary<br>structure               | Boundary<br>catalog;<br>anomaly<br>identification;<br>qualitative<br>framework                                              | Geometric<br>classification<br>of each<br>boundary;<br>quantitative<br>decomposition;<br>test protocols                       |
| MATH-1 | <a href="#">[HOWL-<br/>MATH-1-<br/>2026]</a> | $\beta = \pi/4$ is<br>geometric<br>invariant<br>across nine<br>cross-section<br>domains; $Q =$<br>$F \cdot \beta \cdot d^2 \cdot Z$ | Mathematical<br>framework;<br>proven<br>isomorphism;<br>ellipse<br>generalization;<br>invari-<br>ant/variant<br>separation  | Application to<br>physical<br>boundaries;<br>calibration<br>protocol;<br>scope<br>determination                               |
| PHYS-3 | <a href="#">[HOWL-<br/>PHYS-3-<br/>2026]</a> | G never<br>measured<br>outside Earth's<br>Hill sphere; G<br>disagreement<br>unresolved;<br>L1/L2 test<br>proposed                   | Experimental<br>gap; Hill<br>sphere as<br>testable<br>boundary;<br>L1/L2<br>proposal;<br>running<br>couplings<br>comparison | Geometric<br>structure for<br>G; two $d^2$<br>candidates;<br>depth trend<br>test; HUST<br>internal<br>control                 |
| PHYS-4 | <a href="#">[HOWL-<br/>PHYS-4-<br/>2026]</a> | Test program<br>connecting<br>PHYS-1,<br>MATH-1,<br>PHYS-3                                                                          | —                                                                                                                           | Seven ordered<br>tests; kill<br>switch;<br>calibration-<br>first structure;<br>falsification<br>criteria; scope<br>boundaries |

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## APPENDIX K: KILL SWITCH DETAIL

| Condition            | Tests 0-2<br>All Null                                        | Test 0 Con-<br>tradictory +<br>Tests 1-2<br>Null                                                  | Test 0 Con-<br>tradictory +<br>Test 1 or 2<br>Positive                              | Any of<br>Tests 0-2<br>Positive                             | Multiple<br>Positive               |
|----------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------|
| Interpretation       | Framework<br>does not<br>extend to<br>physical<br>boundaries | Framework<br>wrong for<br>quantum<br>boundaries<br>and not<br>supported<br>for gravita-<br>tional | Framework<br>may apply<br>to gravita-<br>tional but<br>not<br>quantum<br>boundaries | Framework<br>extends to<br>at least one<br>boundary<br>type | Strong<br>support                  |
| Action               | STOP                                                         | STOP                                                                                              | Test 4 only;<br>skip 3, 5, 6                                                        | Proceed;<br>positive<br>result<br>determines<br>priority    | Full<br>program                    |
| MATH-1               | Unaffected<br>— cross-<br>section<br>proof is<br>identity    | Unaffected                                                                                        | Unaffected                                                                          | Extended                                                    | Significantly<br>extended          |
| PHYS-1               | Unaffected<br>—<br>qualitative                               | Unaffected                                                                                        | Partially<br>strength-<br>ened<br>(gravita-<br>tional<br>only)                      | Strengthened                                                | Strongly<br>strength-<br>ened      |
| PHYS-3               | Unaffected<br>— G gap<br>factual                             | Unaffected                                                                                        | Connected<br>to<br>geometric<br>framework                                           | Connected                                                   | Strongly<br>connected              |
| Publication<br>value | Scope<br>boundary is<br>a finding                            | Contradiction<br>is a finding                                                                     | Partial<br>scope deter-<br>mination                                                 | Positive<br>results are<br>findings                         | Research<br>program<br>established |

## APPENDIX L: WHAT EACH TEST RESULT MEANS FOR EACH PRIOR PAPER

| Test Result          | Effect on PHYS-1                                     | Effect on MATH-1                             | Effect on PHYS-3                                          |
|----------------------|------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------|
| Test 0 contradictory | Boundary concept weakened for quantum cases          | Unaffected (cross-section proof is identity) | Unaffected (G gap factual)                                |
| Test 0 informative   | Boundary concept strengthened for quantum cases      | Extended to field-theory scattering          | Unaffected                                                |
| Test 0 redundant     | Unaffected                                           | Consistent but not extended                  | Unaffected                                                |
| Test 1 positive      | Anomaly pattern strengthened — G joins it            | Extended to gravitational measurement        | G disagreement reinterpreted as geometric                 |
| Test 1 null          | Unaffected                                           | Scope: not G geometry                        | Disagreement still unexplained                            |
| Test 2 positive      | Depth-dependent readings confirmed for G             | Extended to depth-dependent measurement      | Boundary-dependent G supported                            |
| Test 2 null          | Unaffected                                           | Unaffected                                   | Depth interpretation not supported at available precision |
| Test 3 positive      | Proton puzzle explained by geometric component       | Extended to proton scattering boundary       | Unaffected                                                |
| Test 3 null          | Proton puzzle has non-geometric source (or resolves) | Scope: not proton boundary                   | Unaffected                                                |
| Test 4 positive      | Major confirmation — boundary effects real for G     | Extended to gravitational boundary crossing  | Central hypothesis confirmed                              |
| Test 4 null          | G consistent across Earth Hill sphere                | Unaffected                                   | Hypothesis not supported at measured precision            |
| Test 5 consistent    | Hubble tension explained by cumulative transit       | Extended to cosmological boundaries          | Unaffected                                                |
| Test 5 inconsistent  | Hubble tension non-geometric                         | Scope: not cosmological                      | Unaffected                                                |
| Test 6 positive      | Dark matter partially geometric boundary effect      | Extended to galactic boundaries              | Unaffected                                                |
| Test 6 null          | Dark matter unaffected                               | Scope: not galactic                          | Unaffected                                                |

| Test Result  | Effect on PHYS-1                                          | Effect on MATH-1                     | Effect on PHYS-3                      |
|--------------|-----------------------------------------------------------|--------------------------------------|---------------------------------------|
| All 0-2 null | Qualitative observations stand; no geometric confirmation | Proven scope confirmed; no extension | G gap factual; unconnected to $\beta$ |

## APPENDIX M: SENSITIVITY ANALYSIS — TEST 2

| Variable          | Range Across Experiments       | Corresponding $\Delta\Phi/c^2$                      | G Disagreement Scale               | Ratio: Disagreement / $\Delta\Phi$ | Implication                                                                    |
|-------------------|--------------------------------|-----------------------------------------------------|------------------------------------|------------------------------------|--------------------------------------------------------------------------------|
| Altitude          | 10 – 500 m                     | $\sim 5 \times 10^{-14}$                            | $\sim 5 \times 10^{-4}$ (relative) | $\sim 10^{10}$                     | Positive result would require enormous boundary effect per unit potential      |
| Latitude          | 30.5°N – 51.5°N                | $\sim 3 \times 10^{-12}$ (centrifugal + oblateness) | $\sim 5 \times 10^{-4}$            | $\sim 10^8$                        | Larger lever arm than altitude but still far below disagreement scale          |
| Local g variation | 9.793 – 9.812 m/s <sup>2</sup> | $\sim 2 \times 10^{-3}$ (relative)                  | $\sim 5 \times 10^{-4}$            | $\sim 0.25$                        | Most promising — g variation is within an order of magnitude of G disagreement |

| Variable                   | Range Across Experiments | Corresponding $\Delta\Phi/c^2$             | G Disagreement Scale    | Ratio: Disagreement / $\Delta\Phi$ | Implication                                                             |
|----------------------------|--------------------------|--------------------------------------------|-------------------------|------------------------------------|-------------------------------------------------------------------------|
| Distance to L1/L2 (Test 4) | 6,371 km vs 1,500,000 km | $\sim 6 \times 10^{-10}$ vs $\sim 10^{-8}$ | $\sim 5 \times 10^{-4}$ | $\sim 10^4$ - $10^6$               | L1/L2 measurement has far greater lever arm than any surface comparison |

**Interpretation:** Test 2 is most likely to detect a signal if the depth effect correlates with local  $g$  rather than with gravitational potential. The variation in local  $g$  across experiment sites ( $\sim 0.2\%$ ) is within an order of magnitude of the  $G$  disagreement ( $\sim 0.05\%$ ). If the depth effect scales with local  $g$ , the test has marginal sensitivity. If it scales with gravitational potential (altitude), the test is underpowered by  $\sim 10$  orders of magnitude.

The test is specified regardless because a null result at the available precision still establishes an upper bound, and because the cost is zero — it is a reanalysis of existing data.

## APPENDIX N: CANDIDATE $d^2$ FOR $G$ — DETAILED ANALYSIS

| Candidate               | Physical Basis                                                | What $d^2$ Represents      | Prediction for Test 1                                                          | Prediction for HUST Control                                                               | Testable With Existing Data? |
|-------------------------|---------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------|
| Test mass cross-section | Gravitational interaction mediated through test mass geometry | Bounding area of test mass | $G$ differs systematically between sphere, cylinder, and flat disk test masses | HUST pair should agree (same test mass) — but they don't; $Z$ must account for difference | Yes                          |

| Candidate                 | Physical Basis                                                        | What $d^2$ Represents                                                          | Prediction for Test 1                                                     | Prediction for HUST Control                                                | Testable With Existing Data?  |
|---------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------|
| Source mass cross-section | Gravitational field shaped by source mass geometry                    | Bounding area of source mass                                                   | G differs based on source mass shape (sphere vs cylinder vs mercury tank) | HUST pair should agree (same source mass) — but they don't; Z must account | Yes                           |
| Source-test configuration | Interaction depends on geometric relationship between source and test | Combined geometric factor (aspect ratio, solid angle, relative positioning)    | G differs based on overall configuration                                  | HUST pair has same configuration → should agree → don't → Z fully accounts | Yes                           |
| Gravitational aperture    | Analogous to Poynting vector through aperture (MATH-1 case 6)         | Effective circular area through which gravitational flux passes between masses | G depends on the effective aperture area at measurement distance          | Aperture same for both HUST methods → should agree → don't → Z accounts    | Partially — requires modeling |

**HUST constraint:** All four candidates predict that the HUST 2018 pair (same masses, same geometry, same location, two methods) should have the same  $\beta \cdot d^2$ . They do. The disagreement (6.67484 vs 6.67349) must reside entirely in Z — the method-specific impedance. This is consistent: Z differs between angular acceleration feedback and time-of-swing because the measurement mechanism differs. The HUST pair confirms that Z is a real and significant factor, independent of geometry.

**Protocol:** Do not pre-select a candidate. Sort existing G data by all geometric parameters simultaneously. Test each for correlation. Let the data identify which parameter, if any, correlates. If none correlates, no candidate is supported.

## APPENDIX O: OPEN QUESTIONS NOT ADDRESSED

| Question                                                               | Why Not Addressed                                                                                                                 | Prerequisite                                                                                           | Where It Belongs                                         |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| What determines the per-boundary transit correction for light?         | Required for Test 5 but not derivable from MATH-1 alone                                                                           | May require new theoretical work connecting $\beta$ to photon-boundary interaction                     | Dedicated theoretical paper; critical blocker for Test 5 |
| Does confinement have a geometric analog in momentum space?            | MATH-1 framework is spatial; momentum-space extension not developed                                                               | Mathematical extension of $\beta$ to non-spatial geometries                                            | Future MATH paper                                        |
| What is the n-dimensional generalization of $\beta$ ?                  | MATH-1 Appendix K notes the sequence $\pi/4, \pi/6, \pi^2/32 \dots$ ; not connected to physical tests                             | Mathematical analysis of dimensional sequence                                                          | Future MATH paper                                        |
| Does the muon g-2 anomaly decompose under the framework?               | Requires identifying the relevant boundary and cross-section for vacuum interactions                                              | Test 0 result ( $\alpha$ calibration); understanding of how $\beta$ applies to loop-level calculations | Future PHYS paper; depends on Test 0                     |
| Are there boundary effects on $\hbar$ , c, or k B?                     | PHYS-3 focuses on G; other constants not analyzed for boundary gaps                                                               | Independent experimental gap analysis for each constant                                                | Future PHYS papers                                       |
| Can the Gibbs overshoot constant relate to the per-transit correction? | Gibbs constant $\approx 0.0895$ ; if per-transit correction is very small, Gibbs is too large; possible higher-order relationship | Mathematical analysis of $\text{Si}(\pi)/\pi - 1/2$ in terms of $\beta$                                | Future MATH paper; speculative                           |
| Does the framework apply to black hole horizons?                       | Event horizons are geometric (spherical) but involve extreme spacetime curvature                                                  | Test 4 result; theoretical extension to strong-field regime                                            | Future PHYS paper                                        |

| Question                                                   | Why Not Addressed                              | Prerequisite                                            | Where It Belongs              |
|------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------|-------------------------------|
| Can existing spacecraft at L1/L2 be repurposed for Test 4? | Engineering question beyond this paper's scope | Assessment of JWST, DSCOVR, Gaia precision capabilities | Engineering feasibility study |

APPENDIX P: SERIES PUBLICATION RECORD

| Paper  | Registry                           | Date       | Domain               | Status   | AI Model             | Key Claim                                                                        |
|--------|------------------------------------|------------|----------------------|----------|----------------------|----------------------------------------------------------------------------------|
| PHYS-1 | <a href="#">[HOWL-PHYS-1-2026]</a> | March 2026 | Foundational Physics | Complete | Claude Opus 4.6      | Mass is inertia; soliton boundaries; three anomaly correlations                  |
| PHYS-2 | <a href="#">[HOWL-PHYS-2-2026]</a> | March 2026 | Foundational Physics | Complete | Claude 4.5<br>Sonnet | Constants run; the word “constant” contradicts the data                          |
| PHYS-3 | <a href="#">[HOWL-PHYS-3-2026]</a> | March 2026 | Foundational Physics | Complete | Claude 4.5<br>Sonnet | G never measured outside Earth Hill sphere                                       |
| MATH-1 | <a href="#">[HOWL-MATH-1-2026]</a> | March 2026 | Mathematics          | Complete | Claude Opus 4.6      | $\beta = \pi /4$ ; nine equations are one; $Q = F \cdot \beta \cdot d^2 \cdot Z$ |

| Paper  | Registry                             | Date       | Domain               | Status   | AI Model        | Key Claim                                                 |
|--------|--------------------------------------|------------|----------------------|----------|-----------------|-----------------------------------------------------------|
| PHYS-4 | [ <a href="#">HOWL-PHYS-4-2026</a> ] | March 2026 | Foundational Physics | Complete | Claude Opus 4.6 | Test program; seven tests; kill switch; calibration first |

END HOWL-PHYS-4-2026

Registry: [[HOWL-PHYS-4-2026](#)]

Status: Complete

Domain: Foundational Physics / Measurement Theory

Central Question: Does  $Q = F \cdot \beta \cdot d^2 \cdot Z$ , proven for geometric cross-sections, apply to coherent pattern boundaries where measurement anomalies are documented?

Method: Classify boundaries by geometry; specify seven tests ordered by achievability; calibrate against known results before applying to unknown cases; commit to kill switch

Key Structure: Three existing-data tests (achievable now), one numerical analysis test (achievable now), one new-instrument test (achievable), one requiring prior theoretical work (underspecified), one far-future; kill switch at Tests 0-2

Foundation: PHYS-1 (boundaries and anomalies), MATH-1 (geometric invariant), PHYS-3 (G experimental gap)

Primary Limitation: The per-boundary transit correction magnitude is unknown; Test 5 (Hubble tension) cannot be executed without it

Falsification: Seven specific criteria plus overall kill switch

APPENDIX Q: THE CALIBRATION CASE —  $\alpha$  RUNNING AS FRAMEWORK VALIDATION

The running of  $\alpha$  is the only case where both the boundary (vacuum polarization cloud) and the readings at multiple depths (collider measurements at multiple energies) are fully characterized by the institution. This table specifies what the  $Q = F \cdot \beta \cdot d^2 \cdot Z$  decomposition must reproduce to pass calibration.



| Energy<br>$\sqrt{s}$<br>(GeV) | $\alpha^{-1}$<br>Mea-<br>sured | Measured<br>Source                       | Published<br>$\sigma$<br>total<br>(nb) | Dominant<br>Pro-<br>cess                                | QED<br>VP<br>Cor-<br>rec-<br>tion<br>$\Delta\alpha$ | QED<br>Ver-<br>tex<br>Cor-<br>rec-<br>tion | Higher-<br>Order<br>QED | Total<br>Pub-<br>lished<br>Z | Framework<br>Must<br>Re-<br>pro-<br>duce<br>Z To               |
|-------------------------------|--------------------------------|------------------------------------------|----------------------------------------|---------------------------------------------------------|-----------------------------------------------------|--------------------------------------------|-------------------------|------------------------------|----------------------------------------------------------------|
| 0<br>(atomic<br>limit)        | 137.036                        | Static<br>mea-<br>sure-<br>ments,<br>QHE | N/A<br>— no<br>scat-<br>tering         | N/A                                                     | 0<br>(defi-<br>ni-<br>tion)                         | 0                                          | 0                       | 1<br>(base-<br>line)         | Exact<br>—<br>this is<br>the cali-<br>bra-<br>tion an-<br>chor |
| 1.02<br>( $\varphi$<br>mass)  | $\sim 136.8$                   | CMD-<br>2,<br>SND<br>(Novosi-<br>birsk)  | $\sim 1500$<br>(hadronic)              | $e^+e^- \rightarrow$<br>hadrons                         | $\sim 0.002$                                        | Small                                      | Negligible              | 1.002                        | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty            |
| 3.097<br>( $J/\psi$ )         | $\sim 135.5$                   | BES<br>(Bei-<br>jing)                    | $\sim 3000$<br>(reso-<br>nance)        | $e^+e^- \rightarrow$<br>$J/\psi \rightarrow$<br>hadrons | $\sim 0.011$                                        | Small                                      | Small                   | $\sim 1.011$                 | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty            |
| 10.58<br>( $\Upsilon(4S)$ )   | $\sim 133.0$                   | BaBar,<br>Belle                          | $\sim 1.1$<br>(con-<br>tin-<br>uum)    | $e^+e^- \rightarrow b\bar{b}$                           | $\sim 0.028$                                        | Moderate                                   | Small                   | $\sim 1.030$                 | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty            |
| 29.0<br>(PEP)                 | $\sim 131.0$                   | MARK<br>II<br>(SLAC)                     | $\sim 0.22$                            | $e^+e^- \rightarrow \mu^+ \mu^-$                        | $\sim 0.034$                                        | Moderate                                   | Small                   | $\sim 1.046$                 | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty            |
| 34.0<br>(PE-<br>TRA)          | $\sim 130.5$                   | TASSO,<br>JADE<br>(DESY)                 | $\sim 0.15$                            | $e^+e^- \rightarrow \mu^+ \mu^-$                        | $\sim 0.039$                                        | Moderate                                   | Small                   | $\sim 1.048$                 | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty            |

| Energy<br>$\sqrt{s}$<br>(GeV) | $\alpha^{-1}$<br>Mea-<br>sured | Published<br>$\sigma$<br>Source                          | total<br>(nb)                | Dominant<br>Pro-<br>cess                     | QED<br>VP<br>Cor-<br>rec-<br>tion<br>$\Delta\alpha$ | QED<br>Ver-<br>tex<br>Cor-<br>rec-<br>tion | Higher-<br>Order<br>QED                                     | Total<br>Pub-<br>lished<br>Z | Framework<br>Must<br>Re-<br>pro-<br>duce<br>Z To                        |
|-------------------------------|--------------------------------|----------------------------------------------------------|------------------------------|----------------------------------------------|-----------------------------------------------------|--------------------------------------------|-------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|
| 57.8<br>(TRIS-<br>TAN)        | $\sim 129.0$                   | AMY,<br>TOPAZ,<br>VENUS                                  | $\sim 0.08$                  | $e^+e^- \rightarrow f \bar{f}$               | $\sim 0.052$                                        | Significant                                | Moderate                                                    | 1.062                        | Within<br>pub-<br>lished<br>QED<br>uncer-<br>tainty                     |
| 91.19<br>(Z<br>pole)          | $127.95 \pm 0.02$              | LEP I<br>(ALEPH,<br>DEL-<br>PHI,<br>L3,<br>OPAL),<br>SLD | $\sim 30$<br>reso-<br>nance) | $e^+e^- \rightarrow Z \rightarrow f \bar{f}$ | $\sim 0.059$                                        | Significant                                | Significant<br>EW<br>cor-<br>rec-<br>tions<br>domi-<br>nate | 1.071<br>(QED<br>only)       | $\pm 0.02$<br>in $\alpha^{-1}$<br>—<br>tight-<br>est<br>con-<br>straint |
| 130-<br>207<br>(LEP<br>II)    | $\sim 127.0$                   | LEP<br>II                                                | $\sim 0.01$ -<br>0.05        | $e^+e^- \rightarrow W^+W^-, f \bar{f}$       | $\sim 0.063$                                        | Significant                                | EW +<br>QCD                                                 | $\sim 1.079$                 | Within<br>pub-<br>lished<br>EW<br>fit<br>uncer-<br>tainty               |

**Calibration protocol:** At each energy, extract  $\beta \cdot d^2(E)$  from the published total cross-section. Compute  $Z(E) = \alpha(E) / [F(E) \cdot \beta \cdot d^2(E)]$ . Compare  $Z(E)$  to the published total QED correction at that energy. Agreement confirms the decomposition is consistent. Disagreement kills the field-theory branch of the program.

**What “informative” means specifically:** If the separation into  $\beta \cdot d^2(E)$  and  $Z(E)$  reveals that the geometric component tracks a smooth monotonic function of energy while the QED corrections produce localized deviations at threshold crossings (charm, bottom, Z pole), the decomposition has identified structure that the total running curve obscures. This would be Outcome B — structurally informative.

## APPENDIX R: G MEASUREMENT GEOMETRIC PARAMETERS — EXTENDED

This table extends Appendix D with the detailed geometric parameters needed for Test 1. Every parameter is from published apparatus descriptions.

| Experiment     | Test Mass Shape    | Test Mass Dimensions                   | Test Mass Bounding Area $d^2$ (m <sup>2</sup> ) | $\beta \cdot d^2$ (m <sup>2</sup> ) | Source Mass Shape  | Source Mass Dimensions           | Separation $r$ (m) | Configuration Geometry      | Solid Angle Subtended  |
|----------------|--------------------|----------------------------------------|-------------------------------------------------|-------------------------------------|--------------------|----------------------------------|--------------------|-----------------------------|------------------------|
| Cavendish 1798 | Sphere             | $d \approx 5$ cm                       | $2.5 \times 10^{-3}$                            | $2.0 \times 10^{-3}$                | Sphere             | $d \approx 30$ cm                | $\sim 0.23$        | Sphere—horizontal           | $\sim 0.13$ sr         |
| NIST 1982      | Cylinder           | $d \approx 2.5$ cm, $h \approx 2.5$ cm | $6.25 \times 10^{-4}$                           | $4.9 \times 10^{-4}$                | Cylinder           | $d \approx 10$ cm                | $\sim 0.10$        | Cylinder—horizontal         | $\sim 0.79$ sr         |
| UWash 2000     | Flat pendulum      | $\sim 8$ cm $\times$ 8 cm              | $6.4 \times 10^{-3}$                            | $5.0 \times 10^{-3}$ (rectangular)  | Sphere array       | 8 spheres, $d \approx 6$ cm each | $\sim 0.10$        | Flat—sphere array, rotating | Complex — multi-source |
| BIPM 2001      | Cylinder           | $d \approx 11$ cm, $h \approx 11$ cm   | $1.21 \times 10^{-2}$                           | $9.5 \times 10^{-3}$                | Cylinder           | $d \approx 11$ cm                | $\sim 0.12$        | Cylinder—cylinder           | $\sim 0.66$ sr         |
| Zurich 2006    | Mass on beam       | $\sim 5$ cm equivalent                 | $\sim 2.5 \times 10^{-3}$                       | $\sim 2.0 \times 10^{-3}$           | Mercury tanks      | $\sim 50$ cm cylindrical         | $\sim 0.30$        | Point—extended source       | $\sim 0.22$ sr         |
| HUST 2009      | Sphere             | $d \approx 3$ cm                       | $9.0 \times 10^{-4}$                            | $7.1 \times 10^{-4}$                | Sphere             | $d \approx 15$ cm                | $\sim 0.08$        | Sphere—sphere               | $\sim 2.76$ sr         |
| LENS 2014      | Cold Rb atom cloud | $\sim 1$ mm effective diameter         | $\sim 10^{-6}$                                  | $\sim 8 \times 10^{-7}$             | Tungsten cylinders | 150 kg each, $\sim 25$ cm        | $\sim 0.26$        | Point-like—extended         | $\sim 0.73$ sr         |

| Experiment | Test Mass Shape | Test Mass Dimensions                 | Test Mass Bounding Area |                                     | Source Mass Shape | Source Mass Dimensions | Separation r (m) | Configuration     | Solid Angle Subtended |
|------------|-----------------|--------------------------------------|-------------------------|-------------------------------------|-------------------|------------------------|------------------|-------------------|-----------------------|
|            |                 |                                      | $d^2$ (m <sup>2</sup> ) | $\beta \cdot d^2$ (m <sup>2</sup> ) |                   |                        |                  |                   |                       |
| BIPM 2014  | Cylinder        | $d \approx 11$ cm, $h \approx 11$ cm | $1.21 \times 10^{-2}$   | $9.5 \times 10^{-3}$                | Cylinder          | $d \approx 11$ cm      | $\sim 0.12$      | Cylinder-cylinder | $\sim 0.66$ sr        |
| HUST 2018  | Sphere          | $d \approx 3$ cm                     | $9.0 \times 10^{-4}$    | $7.1 \times 10^{-4}$                | Sphere            | $d \approx 15$ cm      | $\sim 0.08$      | Sphere-sphere     | $\sim 2.76$ sr        |

**Test 1 protocol:** Sort by test mass bounding area  $\beta \cdot d^2$ . Sort by source mass bounding area. Sort by solid angle subtended (the ratio of source mass cross-section to  $4 \pi r^2$ ). Sort by configuration type (sphere-sphere vs cylinder-cylinder vs point-extended). Compute Spearman rank correlation between G value and each geometric variable. Test for systematic offset between shape categories.

#### Key observations from the raw data:

The LENS experiment has a test mass cross-section  $\sim 10,000 \times$  smaller than other experiments (atom cloud vs macroscopic mass). It also produces the lowest G value in the modern record (6.67191). If test mass cross-section correlates with G, LENS is the extreme case.

The UWash experiment has a flat (non-cylindrical, non-spherical) test mass geometry and uses a rotating configuration. It produces  $G = 6.67422$ , the most precisely measured single value. The unique geometry makes it a natural outlier test — if geometry matters, the non-standard shape should produce a distinctive result.

## APPENDIX S: PROTON RADIUS — QED CORRECTION BREAKDOWN

For Test 3, the residual after QED corrections is the signal. This table itemizes the QED corrections for both electron and muon probes so the residual can be isolated.

| Correction Type                   | Hydrogen (electron probe)                    | Muonic Hydrogen (muon probe)                 | Ratio $\mu / e$ | Physical Origin                                                       |
|-----------------------------------|----------------------------------------------|----------------------------------------------|-----------------|-----------------------------------------------------------------------|
| One-loop VP (electron)            | ~27.13 meV (2S-2P splitting contribution)    | ~205.007 meV                                 | ~7.6            | Virtual $e^+e^-$ pairs; dominant in both                              |
| One-loop VP (muon)                | ~0.017 meV                                   | ~0.017 meV                                   | ~1              | Virtual $\mu^+ \mu^-$ pairs; small in both                            |
| One-loop VP (hadron)              | ~0.044 meV                                   | ~11.36 meV                                   | ~258            | Virtual quark loops; much larger for muonic due to deeper penetration |
| Two-loop VP                       | ~1.51 meV                                    | ~1.51 meV (scaled)                           | ~1              | Higher-order vacuum polarization                                      |
| Self-energy                       | ~-26.90 meV                                  | ~-26.90 meV (scaled by mass ratio)           | Mass-dependent  | Electron/muon self-energy                                             |
| Recoil corrections                | ~0.059 meV                                   | ~0.059 meV (scaled)                          | ~1              | Finite nuclear mass effects                                           |
| Nuclear polarizability            | ~0.015 meV                                   | ~0.015 meV                                   | ~1              | Nuclear structure beyond charge radius                                |
| Proton size (the target)          | ~-1.28 meV $\times r_p^2$ (fm <sup>2</sup> ) | ~-5.23 meV $\times r_p^2$ (fm <sup>2</sup> ) | ~4.1            | Finite proton charge radius — this is what is being measured          |
| Total theoretical (excluding r p) | ~1.92 meV                                    | ~180.5 meV                                   | ~94             | All QED corrections sum                                               |

### Test 3 procedure:

Step 1 — Take the measured 2S-2P transition energy for hydrogen and muonic hydrogen from the most recent published values.

Step 2 — Subtract all published QED corrections except the proton size term. What remains is the proton size contribution.

Step 3 — Extract  $r_p$  from the proton size contribution for each probe independently.

Step 4 — Compute the residual:  $\Delta r_p = r_p(\text{electron}) - r_p(\text{muon})$ .

Step 5 — Compute the probe cross-section ratio:  $\beta \cdot d^2_e / \beta \cdot d^2_\mu = 207^2 = 42,849$ .

Step 6 — Test whether  $\Delta r_p / r_p$  correlates with the cross-section ratio through a function involving  $\beta$ .

**The hadronic VP correction is the key.** It is  $258 \times$  larger for muonic hydrogen because the muon probes  $207 \times$  deeper, where hadronic vacuum polarization contributes more strongly. This correction is the largest source of theoretical uncertainty in the muonic measurement. If the geometric framework identifies a systematic component in the hadronic VP that current QED calculations do not capture, it would appear as a structured residual correlating with the cross-section ratio.

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## APPENDIX T: THE PER-TRANSIT CORRECTION — CONSTRAINTS FROM OBSERVATIONAL DATA

Test 5 requires the per-boundary transit correction magnitude. This appendix derives constraints from the observational data, even though the theoretical derivation is missing.

| Constraint Source                 | Constraint                                                                                                        | Derivation                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Hubble tension ratio              | $x^{N_{\text{eff}}} = 0.923$ where $x$ is per-transit correction and $N_{\text{eff}}$ is effective boundary count | Direct from $H_0(\text{CMB})/H_0(\text{local})$                                                                |
| CMB photon survival               | $x^{N_{\text{eff}}} > 0$ — CMB photons arrive; correction cannot be absorptive                                    | Photons are detected; correction is a shift, not an attenuation                                                |
| CMB blackbody spectrum            | Correction must preserve blackbody shape to $<10^{-5}$ (COBE/FIRAS)                                               | CMB spectrum is the most perfect blackbody measured; any frequency-dependent correction is tightly constrained |
| Spectral line integrity           | Correction must preserve atomic transition frequencies (lines are identified at cosmological distances)           | Spectral lines from $z > 6$ quasars are identified at correct frequency ratios                                 |
| Gravitational lensing consistency | Correction must be consistent with lensing predictions from known mass distributions                              | Lensing magnifications match GR predictions for clusters with independently measured masses                    |
| $N_{\text{eff}}$ Estimate         | Per-Transit $x$                                                                                                   | $\ln(x)$                                                                                                       |
| —                                 | —                                                                                                                 | —                                                                                                              |

| Constraint Source          | Constraint | Derivation             |
|----------------------------|------------|------------------------|
| 50 (major structures only) | 0.99839    | $-1.61 \times 10^{-3}$ |
| 100                        | 0.99920    | $-8.01 \times 10^{-4}$ |
| 500                        | 0.99984    | $-1.60 \times 10^{-4}$ |
| 1,000                      | 0.99992    | $-8.01 \times 10^{-5}$ |
| 5,000                      | 0.999984   | $-1.60 \times 10^{-5}$ |
| 10,000                     | 0.999992   | $-8.01 \times 10^{-6}$ |
| 50,000                     | 0.9999984  | $-1.60 \times 10^{-6}$ |
| 100,000                    | 0.9999992  | $-8.01 \times 10^{-7}$ |

**The correction is small but nonzero.** For any reasonable  $N_{\text{eff}}$  (50 to 100,000), the per-transit correction is between 0.8 ppm and 1,610 ppm — far smaller than  $\beta$  itself (which is  $\sim 0.785$  or a 21.5% reduction) but far larger than zero.

**Structural question:** What determines  $x$ ? Candidates:

| Candidate Expression                 | Value                                                              | Matches Any $N_{\text{eff}}$ ?                                    |
|--------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|
| $\beta$ itself                       | 0.785                                                              | No — far too large; two transits would reduce signal by 38%       |
| $1 - \beta$                          | 0.215                                                              | No — still too large                                              |
| $\beta/N$ for some $N$               | Varies                                                             | Possible — but what determines $N$ ?                              |
| $(1 - 1/\beta^k)$ for some $k$       | Varies                                                             | Parameter fitting, not derivation                                 |
| Ratio of wavelength to boundary size | $\sim 10^{-20}$ to $10^{-6}$ depending on wavelength and structure | Plausible range — submillimeter CMB photon vs Mpc-scale structure |
| $R_4 = \pi^2/32$ (from MATH-5)       | 0.308                                                              | No — still too large as direct correction                         |
| Gravitational potential at boundary  | $\Phi/c^2 \sim 10^{-5}$ to $10^{-6}$ per structure                 | Consistent with $N_{\text{eff}} \sim 1,000$ -10,000 range         |

**The gravitational potential candidate is the most physically motivated.** If the per-transit correction is  $x \approx 1 - \Phi/c^2$  where  $\Phi$  is the gravitational potential at the boundary of each coherent structure, then for typical structures with  $\Phi/c^2 \sim 10^{-5}$ ,  $N_{\text{eff}} \sim 8,000$  structures would produce the observed 0.923 ratio. This is within the plausible range from large-scale structure surveys. The derivation of this candidate from first principles is the missing theoretical piece.

## APPENDIX U: HUST INTERNAL CONTROL — DETAILED ANALYSIS

The HUST 2018 two-method result is the most constraining single data point for the framework. This appendix analyzes it in detail.

| Parameter                      | Method 1 (Angular Acceleration)                                           | Method 2 (Time of Swing)                                                 | Same or Different?   |
|--------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------|
| Test mass                      | Same spheres                                                              | Same spheres                                                             | Same                 |
| Source mass                    | Same spheres                                                              | Same spheres                                                             | Same                 |
| Separation                     | Same                                                                      | Same                                                                     | Same                 |
| Location                       | Same laboratory, Wuhan                                                    | Same laboratory                                                          | Same                 |
| Altitude                       | ~30 m                                                                     | ~30 m                                                                    | Same                 |
| Latitude                       | 30.5°N                                                                    | 30.5°N                                                                   | Same                 |
| Local g                        | 9.7934 m/s <sup>2</sup>                                                   | 9.7934 m/s <sup>2</sup>                                                  | Same                 |
| Temperature                    | Controlled, similar                                                       | Controlled, similar                                                      | Approximately same   |
| Measurement principle          | Servo loop maintains angular position; measures restoring torque          | Free oscillation; measures period change when source masses repositioned | Different            |
| G result ( $\times 10^{-11}$ ) | $6.67484 \pm 0.00012$                                                     | $6.67349 \pm 0.00018$                                                    | Different by 135 ppm |
| $\beta \cdot d^2$              | Identical (same masses)                                                   | Identical                                                                | Same                 |
| Z (method impedance)           | Servo feedback loop, control electronics, calibration of restoring torque | Pendulum fiber properties, air damping, frequency measurement precision  | Different            |

**Framework interpretation:**  $\beta \cdot d^2$  is identical for both methods. The disagreement resides entirely in Z — the method-specific impedance. This confirms that Z is a real and significant factor. The magnitude of the Z-induced disagreement (135 ppm) is comparable to the total spread across all modern experiments (~500 ppm from lowest to highest). This means Z accounts for a significant fraction of the total disagreement even when geometry is held constant.

**Implication for Test 1:** If the HUST pair shows ~135 ppm disagreement from Z alone (geometry identical), then the total ~500 ppm spread across all experiments is the sum of geometric variation plus Z variation. The geometric component can be at most ~365 ppm (500 – 135). If Test 1 finds a geometric correlation at this level, it is consistent. If



the geometric correlation accounts for more than ~365 ppm, something is wrong with the decomposition.

**Implication for Test 4:** The L1/L2 measurement must use a single method to control Z. If both angular acceleration and time-of-swing methods are deployed at L1/L2, the within-method comparison (L1/L2 vs Earth surface, same method) isolates the boundary effect from method-specific Z. The between-method comparison at L1/L2 provides a Z calibration at the boundary.

## APPENDIX V: COMPLETE BOUNDARY TRANSIT TABLE — EARTH SURFACE TO CMB

Every boundary a CMB photon crosses between the surface of last scattering and a ground-based detector, ordered from outermost to innermost.

| Transit # | Boundary                                         | Type                           | Approximate Scale | Gravitational Potential $\Phi/c^2$ | Known Corrections Applied                  | Boundary Effect Modeled?   |
|-----------|--------------------------------------------------|--------------------------------|-------------------|------------------------------------|--------------------------------------------|----------------------------|
| 1         | Surface of last scattering                       | Plasma → neutral transition    | ~46 Gly comoving  | —                                  | Yes — recombination physics, Saha equation | Yes (CMB physics)          |
| 2-100     | Void-filament boundaries (large-scale structure) | Density contrast transitions   | 30-100 Mpc each   | ~10 <sup>-5</sup> each             | Partially — ISW effect, lensing            | Not as coherent boundaries |
| 101-300   | Galaxy cluster halos                             | Dark matter + gas boundary     | 2-5 Mpc each      | ~10 <sup>-5</sup> each             | Partially — SZ effect, lensing             | Not as coherent boundaries |
| 301-1000+ | Individual galaxy halos                          | Dark matter + stellar boundary | 0.1-1 Mpc each    | ~10 <sup>-6</sup> each             | Not individually                           | No                         |

| Transit # | Boundary                 | Type                             | Approximate Scale         | Gravitational Potential $\Phi/c^2$ | Known Corrections Applied                         | Boundary Effect Modeled?         |
|-----------|--------------------------|----------------------------------|---------------------------|------------------------------------|---------------------------------------------------|----------------------------------|
| ~N-10     | Milky Way halo           | Galaxy dark matter halo          | ~300 kpc                  | $\sim 10^{-6}$                     | Partially — foreground subtraction                | Not as boundary transit          |
| ~N-9      | Milky Way disk           | Stellar + gas disk               | ~30 kpc                   | $\sim 10^{-6}$                     | Yes — foreground emission subtraction             | Yes (foreground)                 |
| ~N-8      | Local Bubble wall        | ISM density transition           | ~100 pc                   | $\sim 10^{-8}$                     | Partially — extinction correction                 | Not as boundary                  |
| ~N-7      | Heliosphere              | Solar wind boundary              | ~120 AU                   | $\sim 10^{-8}$                     | Not for CMB                                       | No                               |
| ~N-6      | Earth Hill sphere        | Gravitational dominance boundary | $\sim 1.5 \times 10^6$ km | $\sim 10^{-8}$                     | Not for CMB                                       | No                               |
| ~N-5      | Earth magnetosphere      | Magnetic field boundary          | ~60,000 km                | N/A (magnetic)                     | Charged particle only                             | No (CMB is photons)              |
| ~N-4      | Earth gravitational well | Gravitational potential well     | ~6,400 km depth           | $\sim 7 \times 10^{-10}$           | Yes — gravitational redshift for precision timing | Yes (GR correction)              |
| ~N-3      | Ionosphere               | Plasma boundary                  | ~300-1000 km              | N/A (plasma)                       | Yes — frequency-dependent                         | Yes (radio/microwave correction) |

| Transit # | Boundary                         | Type                 | Approximate Scale | Gravitational Potential $\Phi/c^2$ | Known Corrections Applied                 | Boundary Effect Modeled? |
|-----------|----------------------------------|----------------------|-------------------|------------------------------------|-------------------------------------------|--------------------------|
| ~N-2      | Troposphere/atmospheric boundary | Atmospheric boundary | ~10-50 km         | N/A (refractive)                   | Yes — atmospheric emission and absorption | Yes (atmospheric model)  |
| ~N-1      | Telescope aperture               | Instrument boundary  | ~meters           | N/A                                | Yes — beam pattern, sidelobe correction   | Yes (instrument model)   |
| N         | Detector                         | Measurement boundary | mm                | N/A                                | Yes — detector response calibration       | Yes (detector model)     |

**Modeled vs unmodeled:** The institution models many individual transits — foreground subtraction, atmospheric correction, detector calibration, gravitational redshift, ISW effect, SZ effect, lensing. It does not model the cumulative effect of crossing hundreds of coherent gravitational boundaries as a distinct systematic category. Each transit is handled by its own correction pipeline. The structural question: after all individual corrections, is there a cumulative residual from the boundary crossings themselves?

## APPENDIX W: ROTATION CURVE — FRAMEWORK APPLICATION SKETCH

For Test 6. This is a sketch, not a calculation. The calculation is identified as difficult and dependent on earlier test results.

| Component                 | Standard Model (with dark matter)              | Framework Application (sketch) | Difference |
|---------------------------|------------------------------------------------|--------------------------------|------------|
| Visible mass $M_{vis}(r)$ | Measured from luminosity + mass-to-light ratio | Same                           | None       |

| Component                         | Standard Model<br>(with dark matter)                                                                                    | Framework<br>Application<br>(sketch)                                                                                                           | Difference                                      |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Expected $v(r)$ from visible mass | $v^2 = GM_{\text{vis}}(r)/r \rightarrow$<br>Keplerian decline<br>at large $r$                                           | Same Newtonian<br>calculation                                                                                                                  | None                                            |
| Observed $v(r)$                   | Flat at large $r$                                                                                                       | Same observational<br>data                                                                                                                     | None                                            |
| Discrepancy                       | $v^2_{\text{obs}} - v^2_{\text{vis}} = v^2_{\text{DM}} \rightarrow$ attribute to<br>dark matter halo $M_{\text{DM}}(r)$ | $v^2_{\text{obs}} - v^2_{\text{vis}} = \Delta v^2 \rightarrow$ ask whether<br>geometric<br>boundary<br>correction accounts<br>for any fraction | The question                                    |
| Dark matter halo                  | NFW profile $M_{\text{DM}}(r)$ fitted to<br>match flat curve                                                            | Reduced by<br>whatever fraction<br>the geometric<br>correction accounts<br>for                                                                 | Potentially reduced                             |
| $G$ used                          | Single value from<br>Earth surface                                                                                      | Single value — but<br>question whether $G$<br>at galactic<br>boundary depth<br>differs                                                         | If $G$ differs, $M_{\text{vis}}(r)$ changes too |

**The circular problem:** If  $G$  is boundary-dependent (Test 4), then  $M_{\text{vis}}(r)$  itself changes because  $M$  is derived from luminosity using stellar models that assume Earth-surface  $G$ . The dark matter inference and the  $G$  universality assumption are entangled. Test 6 cannot be cleanly separated from Test 4. A positive Test 4 result changes the input to Test 6.

**The geometric correction:** At each radius  $r$ , the enclosed mass is measured through a gravitational interaction that crosses the galactic boundary structure. The boundary is approximately elliptical.  $\beta$  applies to ellipses at all eccentricities (MATH-1). The question is whether  $\beta$  modifies the effective gravitational coupling at the boundary in a way that changes the inferred enclosed mass.

**What would “positive” look like:** If the geometric correction reduces the dark matter requirement by a specific, calculable fraction — say 10-30% — that is consistent across multiple galaxies with different mass distributions and eccentricities, the framework has explanatory power. If the correction is negligible (<1%) or inconsistent across galaxies, it does not.

**Honest assessment:** This is the weakest test in the program. The galactic boundary is not a clean geometric object. The rectilinear frame is ambiguous at galactic scales. The connection between  $\beta$  and a gravitational coupling correction has not been derived. This

test is included for completeness but should not be the basis for any claim until Tests 0-4 have produced results.

## APPENDIX X: COMPLETE TEST DEPENDENCY GRAPH

| Test                      | Depends On                                              | Provides Input To                                        | Can Run In Parallel With                     | Kill Switch Role          |
|---------------------------|---------------------------------------------------------|----------------------------------------------------------|----------------------------------------------|---------------------------|
| 0 ( $\alpha$ calibration) | None — uses published data only                         | Tests 3, 5, 6 (field-theory branch viability)            | Tests 1, 2                                   | Part of kill switch triad |
| 1 (G geometry)            | None — uses published data only                         | Test 4 (identifies $d^2$ candidate for L1/L2 design)     | Tests 0, 2                                   | Part of kill switch triad |
| 2 (G depth)               | None — uses published data only                         | Test 4 (depth trend constrains expected L1/L2 result)    | Tests 0, 1                                   | Part of kill switch triad |
| 3 (proton radius)         | Test 0 not contradictory                                | Test 5 (constrains per-transit correction)               | Test 4                                       | Second stage              |
| 4 (G at L1/L2)            | Tests 1 or 2 positive (motivates the experiment)        | Tests 5, 6 (if G differs, changes all mass calculations) | Test 3                                       | Second stage              |
| 5 (Hubble tension)        | Per-transit correction derived; Test 3 or 4 positive    | Test 6 (validated transit correction)                    | Test 6 (if per-transit correction available) | Third stage               |
| 6 (rotation curves)       | Test 4 result (G boundary dependence); Test 5 viability | Test 7 (if galactic boundary confirmed)                  | Test 5                                       | Third stage               |
| 7 (solar system boundary) | Strong program support from Tests 0-6                   | None — final test                                        | None                                         | Fourth stage (far future) |

**Critical path:** Tests 0+1+2 (parallel) → Gate → Tests 3+4 (parallel) → Gate → Tests 5+6 (parallel, if supported) → Test 7 (far future). Minimum time to kill switch: however long it takes to reanalyze existing published data for Tests 0, 1, and 2. Estimated: weeks to months, not years.

## APPENDIX Y: WHAT EACH NULL RESULT SPECIFICALLY ESTABLISHES

A null result is not a failure. It is a measurement of zero with an uncertainty bound. This table specifies what each null establishes.

| Test                      | Null Result Statement                                                                                                            | Upper Bound Established                                                                               | Value of the Null                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 0 ( $\alpha$ calibration) | The $Q = F \cdot \beta \cdot d^2 \cdot Z$ decomposition does not reveal structure in QED running beyond what RG already captures | The geometric and QED components are not separable at the level of published $\alpha(E)$ measurements | Confirms QED is complete at this level; framework adds no value for quantum boundaries                       |
| 1 (G geometry)            | G does not correlate with apparatus geometry across published experiments                                                        | Geometric contribution to G disagreement is $< X$ ppm where X depends on statistical power            | The G disagreement is not geometric in origin — narrows the search for its source                            |
| 2 (G depth)               | G does not trend with gravitational potential across published experiments                                                       | Depth effect is $< Y$ ppm per unit of $\Delta\Phi/c^2$                                                | Either no depth effect exists, or it is below the sensitivity of surface measurements (~400m altitude range) |
| 3 (proton radius)         | Proton radius residual does not correlate with probe cross-section ratio                                                         | Geometric component of proton radius puzzle is $< Z$ fm                                               | Puzzle has non-geometric origin; or puzzle resolves through QED improvements                                 |

| Test                | Null Result Statement                                                                       | Upper Bound Established                                                                  | Value of the Null                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 4 (G at L1/L2)      | G at Hill sphere boundary matches Earth surface value within measurement precision          | Boundary effect on G is < measurement uncertainty of the L1/L2 apparatus                 | First positive evidence that G is consistent across Earth's Hill sphere — replaces assumption with measurement                  |
| 5 (Hubble tension)  | Cumulative transit correction does not match $H_0$ ratio                                    | Geometric transit does not explain the Hubble tension at the per-transit correction used | Tension has non-geometric source — or per-transit correction model is wrong                                                     |
| 6 (rotation curves) | $\beta$ correction does not meaningfully reduce dark matter requirement                     | Boundary geometry contributes < W% to rotation curve discrepancy                         | Dark matter problem unaffected by boundary geometry                                                                             |
| All 0-2 null        | The geometric framework does not extend from cross-section equations to physical boundaries | MATH-1 scope confirmed: geometric cross-sections only; not physical boundaries           | <b>The scope boundary is the finding.</b> MATH-1 remains valid within proven scope. The program stops with a documented result. |

**The all-null outcome is publishable.** It establishes that a specific, well-defined extension of a proven mathematical framework to physical boundaries does not produce measurable effects at the precision of available data. This negative result has value — it prevents future researchers from pursuing the same extension without new data or new theoretical motivation.

[[HOWL-PHYS-4-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-4-2026>

[[HOWL-PHYS-1-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-1-2026>

[[HOWL-MATH-1-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/H-MATH-1-2026>

[[HOWL-PHYS-3-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H-PHYS-3-2026>

PHYS-3-2026

[[HOWL-PHYS-2-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/H>  
PHYS-2-2026